

Respirology

Saturday, November 29, 2025

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Royal College Objectives...

- Acute dyspnea
- Chronic dyspnea
- Cough
- Wheeze
- Hemoptysis
- Superior vena cava syndrome
- Interpretation of Pulmonary Function Testing
- Pneumonia
- Chronic obstructive lung disease
- Bronchial asthma
- Interstitial lung disease
- Pulmonary embolism
- Pneumothorax
- Pleural effusion
- Sarcoidosis
- Lung cancer: primary and metastatic including paraneoplastic syndromes

Overview

For each topic will review diagnosis and management, and highlight relevant guidelines and recent evidence

- Airways Disease
 - Asthma
 - COPD
 - Bronchiectasis
- Interstitial Lung Disease
 - IPF
- Pleural Disease
- Sarcoidosis
- Sleep Disordered Breathing
- Pulmonary Hypertension
- Approach to common respiratory problems: Hemoptysis, ABG interpretation/hypoxemia
- BONUS SLIDES: CTS choosing wisely, PFT Primer, Chronic cough, Pulmonary Nodules, Smoking cessation, Important COPD/Asthma trials

PFT Primer & Practice PFTs:
Check out our Online PFT Self-Study Module!

Bonus: Read
on your OWN

Acronyms Used in this Talk

Acronym	Meaning
ATS	American Thoracic Society
AECOPD	Acute Exacerbation of COPD
CTS	Canadian Thoracics Society (respiratoryguidelines.ca)
GINA	G lobal I Nitiative for A sthma
ICS	Inhaled Corticosteroid
LABA	Long acting Beta Agonist (ex. Salmeterol)
LAMA	Long Acting Muscarinic Antagonist (ex. Glycopyrronium, tiotropium)
LVRS	Lung Volume Reduction Surgery
MDD	Multidisciplinary Discussion (as happens in diagnostic w/u of IPF)
NAEB	Non asthmatic eosinophilic bronchitis
NIV	Non Invasive Ventilation (e.g. BiPAP)
NTM	NonTuberculous Mycobacteria
PsA	Pseudomonas Aeruginosa
PFT	Pulmonary Function Test

Asthma

Key resources for Royal College Exam:

1. Global Initiative for Asthma (**GINA**) 2025 guidelines (updated yearly)
 2. Canadian Thoracic Society (**CTS**) Guidelines
 - Recognition and management of severe asthma (2017)
 - Guideline Update: Diagnosis and management of asthma in preschoolers, children and adults (2021)
 - Guideline Update: A focused update on the management of mild and very mild asthma (2021)
- **Major points mostly the same between guidelines except when indicated (if in doubt use CTS)**

MCQ 1 2026

A 28-year-old female who is 16 weeks pregnant is referred to you for suspected asthma. She describes shortness of breath and wheeze triggered by grass cuttings 1-2 times per month. No activity limitations. She does admit to smoking 1 pack of cigarettes a week. She had pre- and post-bronchodilator spirometry and her FEV1 improved by 230mL and 10%. What is the next best step?

- a) Methacholine challenge
- b) Empiric budesonide/formoterol 1-2 inh q4h PRN
- c) Empiric daily low dose fluticasone and PRN salbutamol
- d) Empiric prednisone 40mg PO daily x 5 days

Asthma

Asthma is a **heterogenous** disease characterized by chronic airway inflammation. Symptoms include wheeze, shortness of breath, chest tightness and cough that vary over time and are related to bronchoconstriction, airway wall thickening, increased mucus and **variable expiratory airflow limitation**.

Asthma Diagnosis requires both:

1. History of **variable respiratory symptoms** (e.g. wheeze, SOB, chest tightness, cough) that vary over time and intensity
2. Confirmed **variable expiratory airflow limitation**:

NEED OBJECTIVE TESTING TO HAVE DIAGNOSIS OF ASTHMA

Asthma – Diagnosis (1)

Expiratory airflow limitation:

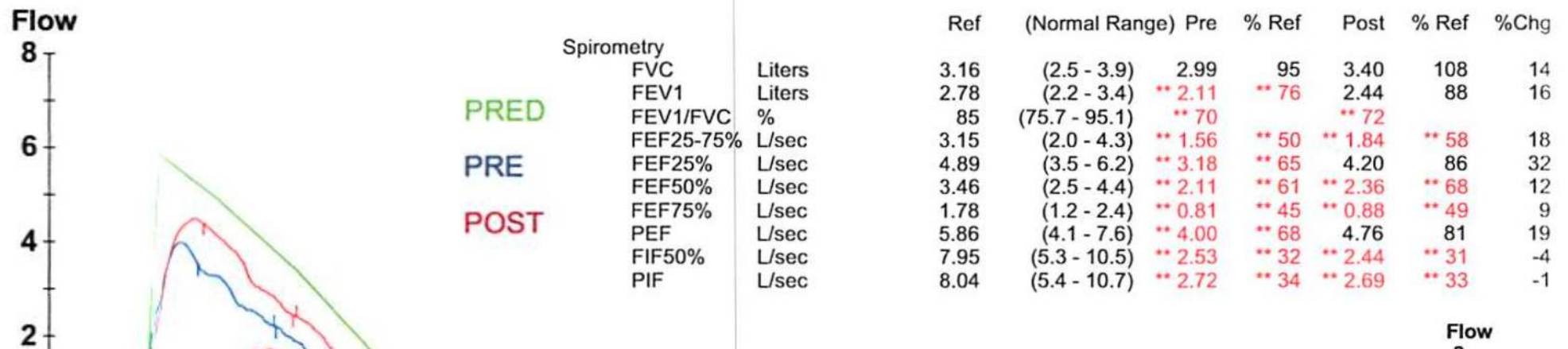
- Since 2024, **GINA no longer requires FEV1/FVC <LLN for diagnosis** – the key is **variable airflow, even if FEV1/FVC is normal range**
- CTS 2021 ‘preferred’ method of diagnosis is FEV1/FVC <LLN AND significant post-bronchodilator response, BUT there are many other ways to diagnose asthma as listed below

Variability, as demonstrated by any of:

- Positive bronchodilator reversibility (10-15 minutes after 200-400 mcg salbutamol)
 - **Improvement in FEV1 by $\geq 12\%$ AND $\geq 200\text{mL}$ post bronchodilator**
- Improvement in lung function with daily ICS-containing treatment x 4 weeks:
 - **Improvement in FEV1 by $\geq 12\%$ AND $\geq 200\text{mL}$ post 4 weeks of ICS treatment**
- Excessive variation in lung function between visits (FEV1 by **12% and 200 cc variation or PEF $\geq 20\%$ where spirometry is not available**)
- Peak Flow Variability – Average daily diurnal PEF variability >8% (CTS - Excessive variability in twice daily PEF over 2 weeks, if doing multiple daily readings >20%)
- Positive Bronchial Challenge Test or Exercise challenge test (e.g. methacholine challenge, see next slides)

FYI: GINA 2025 – role of type 2 biomarkers **FeNO** (fraction exhaled nitric oxide; >50 parts per billion) and **blood eosinophils** (>local ULN): Clinical utility overall still limited, but may help with phenotyping type 2 disease, selection of asthma treatment/predicting response.

Asthma – Diagnosis (1)



Spirometry Interpretation:

FEV1/FVC = 0.7

FEV1 pre bronchodilator = 2.11 L

FEV1 post bronchodilator = 2.44 L

= 330ml and 16% improvement

= asthma!

Asthma – Diagnosis (2)

- Airflow limitation may not be present at the time of initial assessment (ie can have normal spirometry and still have asthma)
 - If normal, may repeat during times of symptoms
 - Can perform methacholine or exercise testing
- Methacholine Challenge – look for drop in FEV1 by $\geq 20\%$
 - measured by PD20 = cumulative dose that causes a 20% drop in FEV1
 - PD20 <100 mcg = POSITIVE
 - PD20 100-400 mcg = borderline
 - PD20 >400 mcg = negative
 - Can also use PC20, the concentration of methacholine that causes drop of 20% in FEV1 – PD20 preferred as more easily standardized.
- Exercise Challenge (Standard Exercise induced bronchoconstriction)
 - Fall in FEV1 of >10% and >200mL from baseline

Assessing Asthma Control –

updated by CTS Nov 2021

- The goal of asthma management is *asthma control*, (includes both asthma symptoms and risk of future adverse outcomes), asthma control should be assessed at each visit
- **Asthma control implies all of the CTS asthma control criteria are present**, see below (NB: This is from the CTS guideline, GINA uses different cutoffs) **KNOW THESE CRITERIA ↓**

Characteristic	Well controlled asthma criteria
Daytime symptoms	≤ 2 d/week
Nighttime symptoms	< 1 night/ week and mild
Physical activity	Normal
Exacerbations	Mild (not requiring systemic steroids or ED visit) and infrequent
Absence from work/school due to exacerbation	None
Need for a reliever (SABA or bud/fom)	≤ 2 doses per week
FEV1 or PEF	≥ 90% of personal best
PEF diurnal variation	<10-15%
Sputum eosinophils	<2-3%

Question: “Does need for PRN inhaler prior to exercise count as poor control?”
Answer: (consensus @ IMR) Those who manage symptoms with prophylaxis alone (eg not as a RELIEVER post exercise) have good control.



Treating asthma: CTS 2021 Key Principles

With each encounter, answer 3 questions:

1. **Is asthma controlled** by CTS criteria?
2. **Are they at higher risk of exacerbations** (Box 1)?
3. **Are they using their inhaler properly?** Are there uncontrolled comorbidities?

First visit: uncontrolled or higher risk of exacerbation
→ daily ICS + PRN SABA

Subsequent visits: If uncontrolled, increase ICS dose in a stepwise fashion and/or add additional controllers: LABA > LAMA or LTRA

BOX 1: Higher Risk of Exacerbation = any 1 of:

- Any history of a previous severe asthma exacerbation (Box 2)
- Poorly controlled asthma per CTS criteria
- Overuse of SABA: more than 2 canisters/yr
- Current smoker

BOX 2: Severe Asthma Exacerbation = any 1 of:

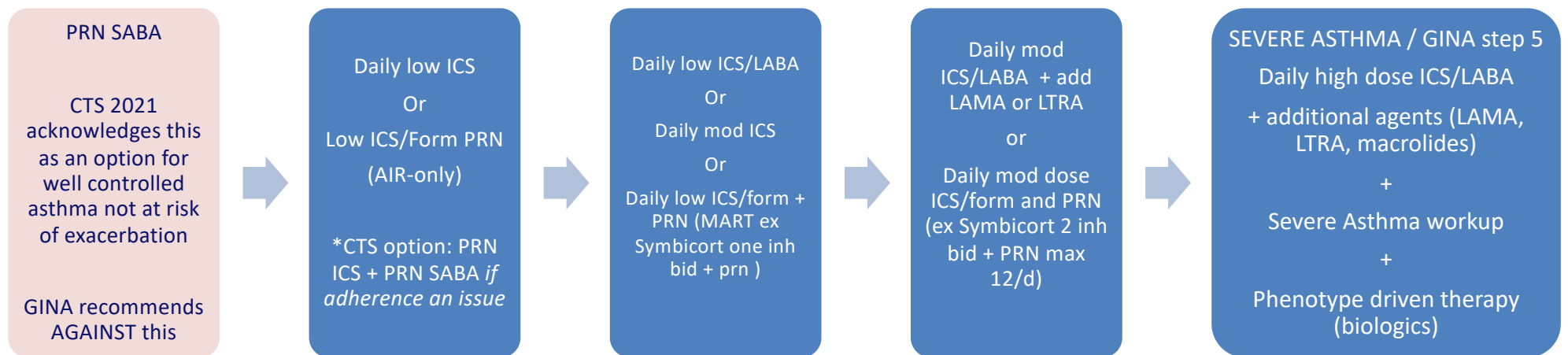
- Requiring systemic steroids
- Requiring ED visit
- Requiring hospital admission

Mild exacerbation = 0/3 above criteria

Asthma treatment

ICS dose in mcg for MCQs (total daily dose)
Fluticasone propionate/budesonide/mometasone
 Low: ≤ 250 /400/200 mcg/d
 Medium: 251-500/400-800/200-400 mcg/d
 High: >500 />800/>400 mcg/d

- Use CTS guidelines over GINA on the exam
- **CTS controller:** Daily ICS (i.e. rather than bud/form or LTRA)
- **CTS reliever:** PRN SABA if on daily ICS. PRN bud/form only if also on maintenance bud/form.
- If a patient is needing high dose ICS and second controller or maintenance OCS, work up for severe asthma
 - GINA discourages routine high dose ICS. Monitoring for systemic adverse effects of inhaled corticosteroids for those on high dose.
 - FYI: Beclomethasone/budesonide have high systemic GC effects. Fluticasone/mometasone/ciclesonide have low systemic GC effects.



*AIR – as needed reliever; MART – maintenance and reliever therapy = GINA terms

Bottom line – GINA and CTS agree: Backbone of asthma treatment is ICS. If uncontrolled, increase ICS dose in a stepwise fashion, continue LABA, and add 3rd controller as needed: LAMA or LTRA – phenotype, biologics, and macrolides if severe

Asthma Treatment – GINA 2025 and CTS 2021

- SABA no longer recommended as monotherapy for mild asthma*
 - Cornerstone of therapy 50yrs ago when asthma thought of as disease of bronchoconstriction
 - But asthma is inflammation – “constant fire in the lungs!”
 - SABA PRN as monotherapy:
 - ↑ risk of exacerbation, ↓ lung function
 - Regular use causes ↑ airway inflammation
 - Over-use associated with ↑ **severe exacerbations and asthma-related death**, even if few interval asthma symptoms!
- **Formoterol-containing ICS^{\$}** recommended due to fast onset of action compared to other LABAs
 - **PRN ICS-formoterol (AIR) in mild asthma** (Cochrane review, 2021):
 - ↓**Symptoms, Exacerbations, Asthma-Related Hospitalization vs SABA alone**
 - ↔severe exacerbations(similar rates) vs daily ICS+SABA prn

*exception in CTS 2021: Pt w/ well controlled asthma, no risk for exacerbation, who prefers to stay on PRN SABA for affordability.
\$ Costly – in Ontario limited use code for ICS-formoterol require failure of ICS



Asthma Study Tips



When studying, stick to CTS

Have a good approach to managing patients with refractory symptoms (eg) for SEVERE ASTHMA:

1. Confirm diagnosis – history, PFT
2. Assess inhaler technique and adherence
3. Assess triggers
4. R/O comorbidities: rhinosinusitis, GERD, vocal cord dysfxn, anxiety, depression, CF, bronchiectasis, ABPA, eGPA, non-tuberculous mycobacteria, immunodeficiency...
5. Consider bronchoscopy if query upper airway disease or atypical infection
6. Addition of additional agents: LAMA, LTRA, macrolides
7. Phenotyping for selection of biologics
8. Consider referral to airways center +/- bronchial thermoplasty

The following BONUS slides are detailed summaries of GINA and CTS, for your quick read, but don't memorize / reference only.

Bonus: Read
on your OWN

Treating asthma: deciphering CTS vs GINA

Key difference #1:

- CTS favours an **ICS controller** and **SABA reliever**
 - Reliever bud/form monotherapy is second line to daily ICS for very mild asthma
 - **Recommended second controller is any “LABA”** (which includes formoterol!)
- GINA favours **ICS/Form for all – it is used as both a controller and reliever**
 - **Recommended second controller is formoterol specifically**, which is a type of LABA that is fast acting (“FABA” - “Track 1”)
 - ICS/non-formoterol LABA is second line (“Track 2”)

Key difference #2:

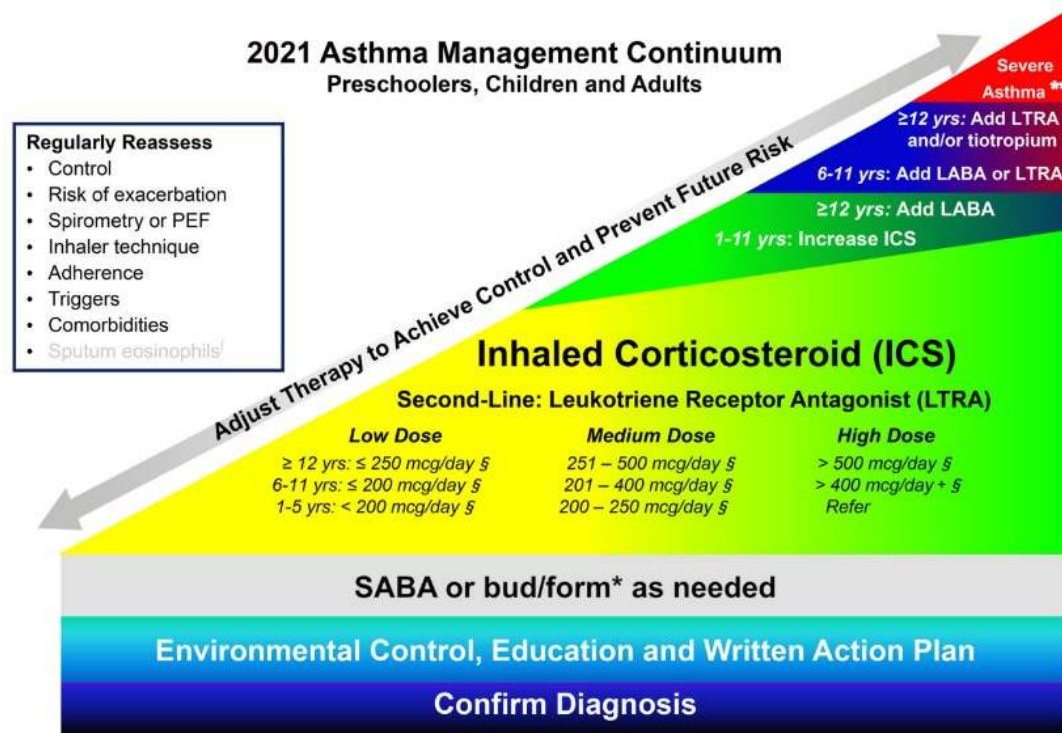
- **CTS accepts PRN SABA** only as an alternative to the **first line daily low ICS option** for those with:
(1) well controlled, (2) not at high risk for exacerbations, (3) pt priority is cost/affordability > asthma control/reducing exacerbations
- **GINA recommends against PRN SABA** as monotherapy in all circumstances

Key difference #3:

- CTS categorizes asthma retrospectively based on therapy required to achieve control (e.g., Mild asthma = controlled with low ICS)
- GINA categorizes asthma based on current symptom burden, lung function, and exacerbations (e.g., Step 4 = daily symptoms)

Bonus: Read
on your OWN

CTS Management Graphics



* Or an alternative ICS/form preparation if another is approved for use as a reliever in the future. Bud/form is approved as a reliever for ≥12 years of age and should only be used as a reliever in individuals using it as monotherapy or in conjunction with bud/form maintenance therapy

§ HFA Fluticasone propionate or equivalent

† Not approved for use in Canada

‡ In adults, 18 years of age and over with moderate to severe asthma assessed in specialist centres

** For severe asthma refer to CTS 2017 Recognition and Management of Severe Asthma Position Statement

Bonus: Read
on your OWN

Asthma Treatment – GINA 2025

New GINA 2025:
“high dose” ICS
removed as routine
option due to adverse
effects of long-term
high dose ICS

MAJOR CHANGE

- Prefer Track 1: No more SABA-only treatment as reliever
- ICS-Formoterol new preferred reliever!
- Symbicort – budesonide formoterol (Step 1-2 per guideline)
- Zenhale – mometasone/formoterol
- Fostair – beclomethasone/formoterol

**Never LABA
monotherapy –
increased risk of
death!**

TRACK 1 Preferred controller and reliever

STEP 1 and 2
PRN low-dose ICS formoterol

STEP 3
Low dose
maintenance ICS-
formoterol

STEP 4
Medium dose
maintenance ICS-
formoterol

STEP 5
Refer for phenotypic
assessment, LAMA
add on +/- Anti IgE,
Anti IL5, Anti IL4,
Anti-TSLP. Consider
high dose ICS
Formoterol.

PREFERRED RELIEVER: prn LOW DOSE ICS FORMOTEROL (Maintenance and Reliever = MART)

TRACK 2 Alternative controller and reliever

STEP 1
Take ICS with SABA PRN

STEP 2
Low dose maintenance
ICS

STEP 3
Low dose
maintenance ICS
LABA

STEP 4
Medium dose
maintenance ICS-
LABA

STEP 5
LAMA (Tiotropium)
+/- refer for
phenotypic
assessment, Anti IgE
Anti IL5, Anti IL4.
Consider high dose
ICS LABA.

ALTERNATIVE (aka not preferred) RELIEVER: AS NEEDED ICS-SABA or AS NEEDED SABA

Asthma Treatment (GINA and CTS)

Pharmacological Pearls:

*If patient has symptom control for 2 months and low risk of exacerbations consider **stepping down therapy***

Before STEP UP Therapy: assess and confirm

- Inhaler technique and adherence
- Comorbidities affecting treatment
- All of the Non-Pharm management

Non-Pharmacological

- Confirm Diagnosis, Educate, Written Asthma Action Plan
- Weight loss, exercise training
- Allergen / trigger avoidance / allergen immunotherapy
- Smoking cessation
- Vaccinations
- Avoid NSAIDs (and maybe beta blockers)
- Comorbidities (GERD, PND, Obesity)

Asthma Treatment – LTRA

- **LTRA: may be appropriate as initial controller unwilling/intolerant of ICS, but less effective than ICS at preventing exacerbations (GINA)**
 - Most effective in aspirin-exacerbated asthma, exercise-induced symptoms, allergic rhinitis
 - **FDA Black Box Warning:** increased suicidality in adolescents and adults

For Adults >12y not achieving asthma control on maintenance low dose ICS (CTS):

- **LABA-ICS is superior to ICS-LTRA**
- ↓Exacerbations and Symptoms, ↑QOL and ↑PFT



Question: “For patient with Samter’s Triad (ASA allergy, asthma, nasal polyps) whose asthma is not well controlled on low dose ICS – would you add LTRA or increase ICS dose?”

Answer: Offer LTRA given ASA-exacerbated asthma

Uncontrolled vs Severe Asthma

(CTS 2017, GINA 2025)

Uncontrolled Asthma:

- Poor symptom control **OR**:
- 1 severe exacerbation/year needing oral steroids
- 1 serious exacerbation requiring **hospital/ICU/ventilation**
- FEV₁ <80% of personal best
- Often due to **modifiable factors**:
 - Poor inhaler technique (~80%)
 - Non-adherence
 - Misdiagnosis (e.g., EGPA, ABPA, vocal cord dysfunction)
 - Comorbidities (GERD, obesity, OSA)
 - Environmental triggers (esp. **tobacco smoke**)

SEVERE Asthma

- Requires **high-dose ICS + second controller** for ≥12 months **OR** Systemic steroids ≥50% of the year **OR** still uncontrolled despite this treatment

WORKUP :

- Investigate for ABPA (see asthma plus slide), other airway conditions (E.g., infection, bronchiectasis, airway stenosis)
- **GINA 2025 - assess biomarkers of type 2 inflammation:** peripheral eosinophil count, sputum eosinophils, and FeNO
 - Eosinophils >0.3 – consider non-asthma causes including strongyloides serology before systemic steroids [GINA 2023]
 - Eosinophils >1.5 – consider investigate for conditions such as EGPA [GINA 2023]
- Consider screening for adverse effects of OCS/high dose ICS (e.g., DEXA scan, a.m. serum cortisol) [GINA 2022]
- Refer to a specialist if possible

Severe Asthma Management (CTS 2017)

(Reaffirmed in CTS 2021; Difficult-to-treat & Severe Asthma GINA 2025)

TREATMENT (akin to STEP 5 on GINA algorithm):

- **LAMA / tiotropium mist inhaler** for uncontrolled asthma despite ICS/LABA
 - Increases time to first severe exacerbation (NEJM 2012)
 - Pts should be on at least medium dose ICS/LABA before considering add on
- **Macrolides** for uncontrolled asthma despite ICS/LABA
- **Biologics** (see next slide) - just know they exist
- Low dose **oral corticosteroids** (absolute last resort – typically a bridge to biologics or suggests undiagnosed asthma plus disease)
- **Bronchial thermoplasty**: role remains unclear, only to be practiced in specialized centers



GINA cautions against routine use of high dose ICS more than 3-6 months.

Biologic Overview (don't memorize)

1. Anti-IgE (Omalizumab)

- Indications: allergic asthma IgE 30 – 700, sensitive to at least 1 perennial allergen, severe despite high dose ICS and one other controller (CTS 2017)
- ↓ exacerbation, ↓ hospitalization, ↓ in ICS dose

2. IL-5 (mepolizumab, reslizumab, benralizumab)

- Indication **severe eosinophilic asthma (generally >300)** and recurrent exacerbation despite high-dose ICS and one other controller (CTS 2017)

Mepolizumab ↓ exacerbation, ↓ in ICS dose, ↓ symptoms and ↑ FEV1 (NEJM 2017)

- ## 3. IL-4/IL-13 (Dupilumab)
- add-on option for severe eosinophilic asthma or those with nasal polyposis or moderate-severe atopic dermatitis. ↓ exacerbation, ↓ in ICS dose and ↑ lung function in patients with **severe eosinophilic asthma**

- ## 4. Anti TSLP (Tezepelumab)
- add on for severe asthma (including non-allergic). TSLP is a cytokine.
↓ exacerbation, ↑ FEV1 and QOL lung function

For **severe** asthma:
If high IgE and allergies: think about omalizumab.
If high eosinophil count, think about all the other biologics.
*Tezepelumab does not require any biomarkers to prescribe

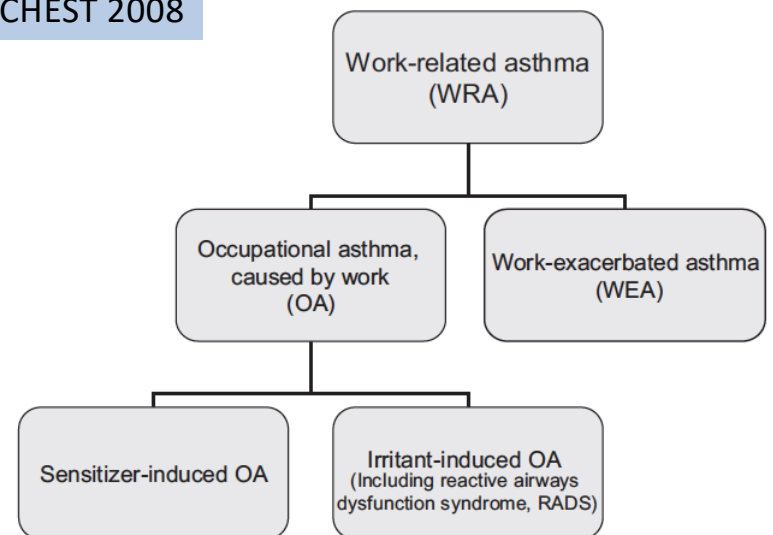
Special Populations (GINA)

- **Seasonal allergic asthma:** start ICS immediately when symptoms commence, and continue for four weeks after relevant pollen season ends
- **Pregnancy:** rule of thirds (1/3 better, 1/3 worse, 1/3 same), exacerbations more common, increased risk of preeclampsia, preterm birth, low birth weight, perinatal mortality. Treat as you would anyone else, do not stop ICS, most evidence for budesonide of all ICS (though likely all safe), **do not withhold oral steroids if exacerbating**. Little data on biologics, though registry data suggests omalizumab safe.
- **ASA exacerbated respiratory disease (AERD / Samter's triad:** asthma, nasal polyps, ASA/NSAID sensitivity): **avoid ASA/NSAIDs (COX-1 inhibitors)**, ICS are mainstay as with other asthma but **usually good response to LTRA**, desensitize to ASA if needed

Work Related Asthma

- Occupational asthma: asthma caused by exposure to stimuli in the workplace
 - Sensitizer-induced asthma (latency of months to years before disease development and risk increases with dose and duration of exposure)
 - Reactive airways dysfunction (RADS) – acute onset after single high-level exposure to vapors, gas, or fumes (ie. Chlorine, bleach) that persists at least 3 months, *treat like asthma exacerbation*
- Work exacerbated asthma – pre-existing asthma worse at work

CHEST 2008



These groupings are not mutually exclusive; e.g. OA can be followed by WEA

Question: How can we differentiate RADS from Vocal Cord Dysfunction that might also be triggered by irritants?

Answer: Clinical scenario, + testing – some RADS may have a normal spirometry but virtually all will have abnormal methacholine challenge.

Asthma 'Plus' Syndromes/Asthma mimics

- **Allergic bronchopulmonary aspergillosis (ABPA)**

- Almost exclusively seen in asthma or CF
- Recurrent exacerbations, fever, brown sputum with 'casts'
- **Treatment: prednisone +/- itraconazole**
- **ISHAM 2024** (FYI only): biologics for steroid dependent ABPA

- **Exercise induced bronchoconstriction**

- Asthma that develops only during activity
- Should be confirmed with objective testing – fall in FEV1 >10%, 200mL
- Regular treatment with ICS avoids EIB, as does regular training and warm up. In patients with mild asthma, options include PRN ICS-form, PRN ICS-SABA, and PRN SABA prior to exercise.

NEW! ISHAM 2024 guidelines

ABPA diagnostic criteria:

1. Predisposing condition (asthma, CF)
2. Serum IgE >500
3. *A. fumigatus*-sp. IgE ≥ 0.35 or skin test
4. Two of three following minor criteria:
 - Positive *A. fumigatus* IgG
 - Consistent imaging: **mucus plugging/bronchiectasis on HRCT** or fleeting opacities on CXR
 - Blood eosinophils >500

Asthma 'Plus' Syndromes / Asthma mimics

- **Bronchiectasis** including Cystic Fibrosis (see Bronchiectasis Section below)
- **Eosinophilic granulomatosis with polyangiitis** (EGPA/Churg Strauss)
 - Asthma, eosinophilia, granulomatous vasculitis (cardiac, sinusitis, allergic rhinitis, transient pulmonary infiltrates, purpura, neurologic, GI), 30-60% have positive p-anca
 - Treatment: prednisone/immunosuppressants – biologics (mepolizumab)
- **Vocal cord dysfunction**
 - Abrupt onset inspiratory stridor, may be misdiagnosed as asthma or complicate asthma
 - Dx via laryngoscopy with adduction of the vocal cords upon inspiration
 - Rx: education, behavior modification, speech therapy, treat GERD

Management of Asthma Exacerbation (GINA)

- Self-management: written asthma action plan
 - May include increased anti-inflammatory reliever
 - May include prescription for OCS. Instructions on when to commence
- Primary care: Severe or life-threatening: transfer to acute care
 - Pillars: rapid-acting bronchodilators (salbutamol), systemic corticosteroids, oxygen supplementation
 - Antibiotics not recommended.
- ED or Acute Care: Life-threatening: consult ICU, prepare to intubate
 - SABA, Atrovent, Oxygen, Steroids
 - Consider IV magnesium 2g, high dose IV corticosteroids (ex methylprednisolone 40-80 mg IV q 8h)
 - Avoid sedation. Theophylline and antibiotics not recommended.



Q) What PEF is safe for discharge?

A) **PEF 60-80% pred** or of personal best and improved symptoms per GINA

Severe: talks in words, agitated, sit hunched forward, RR>30, accessory muscle use, HR>120, O2<90% RA, PEF<50%
Life-threatening: Drowsiness, confusion, silent chest

MCQ 1 2026

A 28-year-old female who is 16 weeks pregnant is referred to you for suspected asthma. She describes shortness of breath and wheeze triggered by grass cuttings 1-2 times per month. No activity limitations. She does admit to smoking 1 pack of cigarettes a week. She had pre- and post-bronchodilator spirometry and her FEV1 improved by 230mL and 10%. What is the next best step?

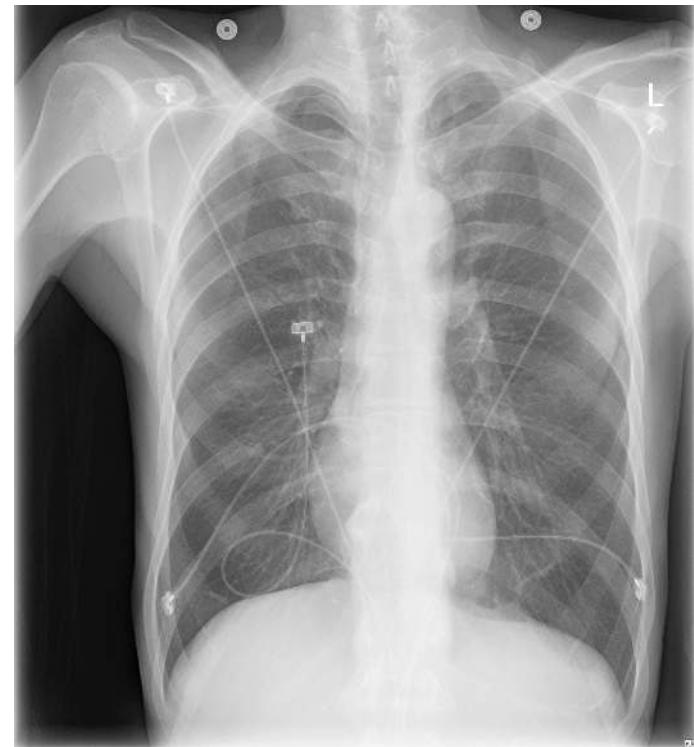
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- b) Empiric budesonide/formoterol 1-2 inh q4h PRN
- c) Empiric daily low dose fluticasone and PRN salbutamol
- d) Empiric prednisone 40mg PO daily

Pregnancy is a contraindication to methacholine challenge. Safest to treat her as presumed asthma since she has a consistent clinical history establish diagnosis objectively post partum. While her daytime symptoms are not frequent enough be considered uncontrolled by CTS criteria, active smoking puts her at high risk of exacerbations. 1st line therapy by CTS guidelines is low ICS/PRN SABA.

COPD

Go to resources for RC:

- 1) **GOLD 2025 (updated yearly)**
- 2) CTS Guidelines:
 - i. **2023 Canadian Thoracic Society Guideline on Pharmacotherapy in Patients with Stable COPD**
 - ii. CTS CPG Update on Pharmacological Therapy of COPD 2019
- 3) ATS CPG Initiating Pharmacologic Treatment in Tobacco-Dependent Adults (2020)



MCQ 2 2026

A 52-year-old man with depression and schizophrenia who has been smoking 1-2 packs per day since he was 12 years old presents to your clinic with a new pulmonary nodule. At the end of your appointment, you speak with him about smoking cessation however, he is reluctant because he does not think he will be successful. He tried nicotine patches 5 years ago and relapsed quickly. Which of the following is the next evidence-based step?

- A. Counsel patient on the risks of ongoing smoking and revisit smoking cessation at each follow up visit
- B. Start varenicline monotherapy.
- C. Start bupropion monotherapy.
- D. Start a non-nicotine containing e-cigarette.

COPD- Diagnosis

CTS 2007, GOLD 2025

Diagnosis: Spirometry is required to make the diagnosis, with a **post-bronchodilator FEV₁/FVC<0.70**

Severity of Airflow Limitation in COPD:

Based on **post-bronchodilator FEV₁**:

- Mild (GOLD 1): $FEV_1 \geq 80\%$ predicted
- Moderate (GOLD 2): $50\% \leq FEV_1 < 80\%$ predicted
- Severe (GOLD 3): $30\% \leq FEV_1 < 50\%$ predicted
- Very Severe (GOLD 4): $FEV_1 < 30\%$ predicted

Proposed “Etiotypes” in 2025 GOLD guidelines

COPD-P: (from pollution exposure) in those exposed to smoke or biomass fuel

COPD-A: COPD and asthma (particularly childhood asthma)

COPD-D: COPD from abnormal lung development

COPD-G : Genetic (AAT)

COPD-C : Cigarettes / Due to smoking

COPD-I : Infection / due to infections (eg Tb, HIV)

COPD-U: COPD from unknown cause

Test all with COPD for AATD
Per NEW 2025 CTS AATD guideline

Who to screen for AATD per CTS 2025:

1. COPD - at time of diagnosis
2. Adult-onset asthma with persistent airway obstruction
3. Unexplained bronchiectasis

COPD Assessment and Management

- Determine disease severity: Airflow limitation via post bronchodilator FEV1
- Impact on patients' health status: mMRC and CAT (COPD Assessment Test) scores
- Labs: Alpha Anti Trypsin ONCE for all
- Imaging: GOLD 2025 - CT thorax if
 - Lung Cancer screening criteria (see med onc lecture)
 - Frequent exacerbations (rule out bronchiectasis or atypical infection), symptoms out of proportion of lung fxn testing
 - Lung volume reduction surgery might be helpful
- Assess Risk of future events: Exacerbations, hospitalizations, death

Goals of management

Prevent disease progression (smoking cessation)

Treat exacerbations and complications of the disease

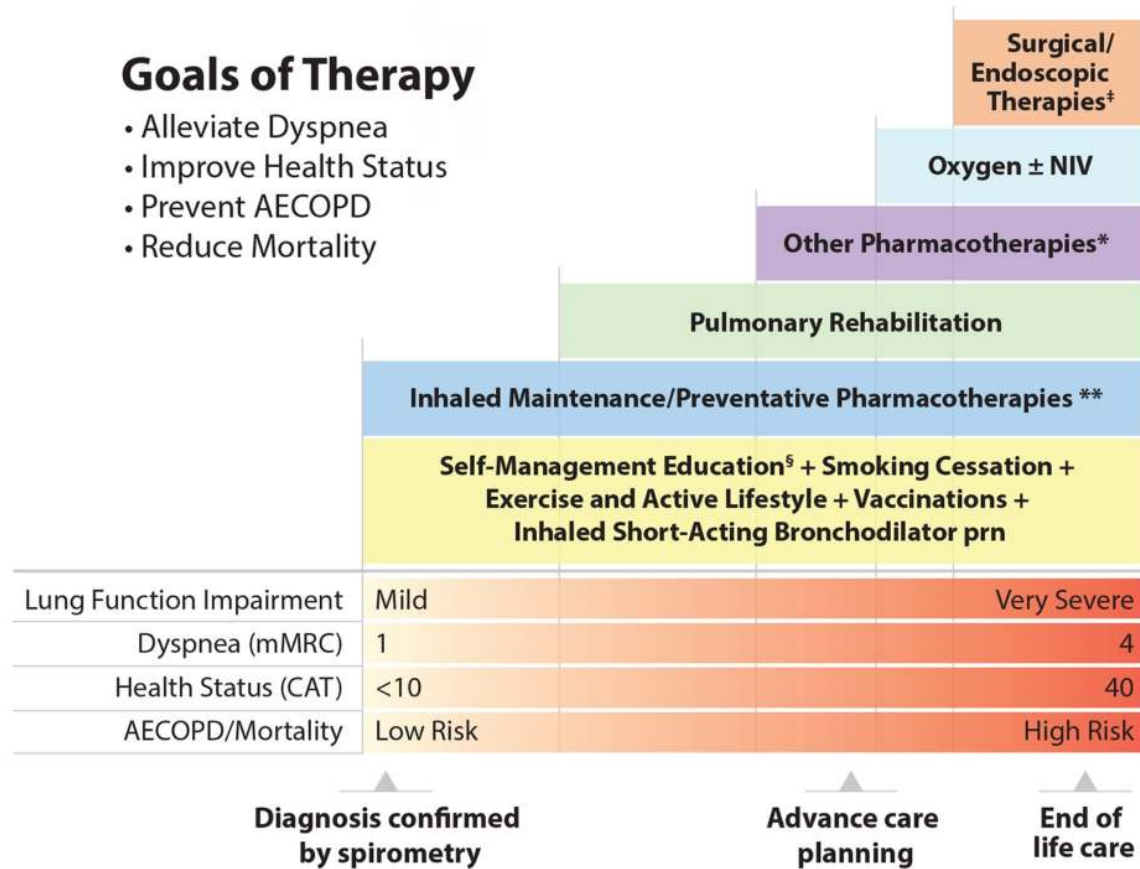
↑exercise tolerance and daily activity, health status

↓Mortality, frequency & severity of AECOPD

CTS 2023 COPD Management

Goals of Therapy

- Alleviate Dyspnea
- Improve Health Status
- Prevent AECOPD
- Reduce Mortality



GOLD/CTS : Non-Pharm Mgmt

- Non-Pharmacologic treatment:
 - **smoking cessation*** and live in a smoke/pollution-free environment
 - **pulmonary rehabilitation*** and remain physically active
 - **supplemental oxygen***
 - vaccination
 - self-management and education
 - review inhaler technique
 - end of life care (palliation, dyspnea management)

***Improve survival in certain subsets of COPD**

GOLD 2024 – Non Pharm & Pharm Treatments with Survival Benefit (CTS 2021 and GOLD 2025)

- **NIV** improves mortality compared to placebo in advanced COPD
 - Stable severe COPD (FEV1 <50%) + PaCO2 ≥ 52
 - Reduced 1 year mortality in small RCT¹, and meta-analysis²
- **LAMA/LABA/ICS** compared to LAMA/LABA relative risk reduction of mortality
 - ETHOS Trial HR 0.51 | IMPACT Trial HR 0.72
 - Population: Symptomatic patients (CAT ≥10) with ≥ 2 moderate or ≥ 1 severe exacerbation

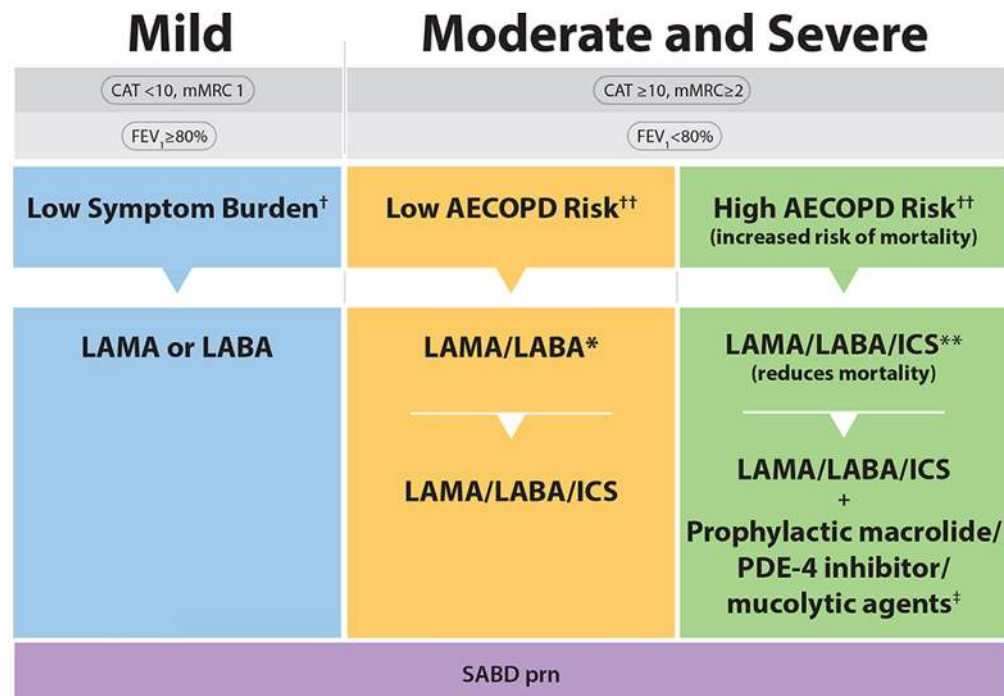
1 Lancet Respir Med. 2014 Sep;2(9):698-705.

2 Tuberc Respir Dis (Seoul). 2022 Jan;85(1):47-55

CTS Stable COPD Management: 2023 Update

Grade	Description – Modified MRC Dyspnea Scale
0	Not troubled by breathlessness except on strenuous exercise
1	Shortness of breath when hurrying on level or walking up slight hill
2	Walks slower than people of the same age on the level b/c of breathlessness or has to stop for breath when walking at own pace on the level
3	Stops for breath after walking about 100m or after a few minutes on the level
4	Too breathless to leave the house or breathless when dressing or undressing

Treatment is based on subjective level of dyspnea and rate of exacerbation, NOT lung function. High risk for AECOPD = ≥ 2 exacerbation in the past year or ≥ 1 requiring hospitalization



CAT = COPD assessment test (Questionnaire)
mMRC = modified Medical Research Council Dyspnea Grade

Stable COPD Management: CTS 2023

Drug	Symptom Burden	AECOPD Risk	FEV1	Benefits	Prescribing Notes
LAMA or LABA Monotherapy	LOW (CAT <10, mMRC≤1)	LOW	≥ 80	↓SOB ↑Exercise ↑Health status vs. placebo	LOW Risk of AECOPD: ≤1 moderate exacerbation in last year (rx abx or pred, no ED/hosp) High Risk of AECOPD: ≥2 moderate or ≥1 severe exacerbation in past year requiring ED visit/hospitalization.
LAMA/LABA Dual	MODERATE (CAT≥10, mMRC≥2)	LOW	< 80	↓SOB ↑Exercise ↑Health status vs. monotherapy LAMA or LABA	Start as initial maintenance if moderate symptoms. If Pt has Asthma also can use ICS/LABA instead here **Step up to LAMA/LABA/ICS if still has moderate symptoms despite dual therapy
LAMA/LABA/ICS	MODERATE to HIGH (CAT≥10, mMRC≥2)	HIGH	< 80	↓ AECOPD Compared to LAMA monotherapy, ICS/LABA and LABA/LAMA ↓Mortality vs LABA/LAMA (and maybe vs ICS/LABA)	Recommend against step down to LAMA/LABA – withdrawing ICS may lower health status/lung fxn. **Unlike GOLD, CTS guideline does not require high eosinophils to initiate LAMA/LABA/ICS

Stable COPD: Beyond Inhalers CTS 2023

Drug	Symptom Burden	AECOPD Risk	FEV1	Benefits	Prescribing Notes
ICS Monotherapy	Don't do it				
Theophylline	Don't do it				
Macrolide	MOD to HIGH CAT $\geq$ 10, mMRC $\geq$ 2)	HIGH	<80	In Addition to LAMA/LABA/ICS: ↓ AECOPD	Add on if pt continues to exacerbate despite inhalers. Risk of hearing impairment, abx resistance, QT prolongation must be weighed vs benefit
Roflumilast or N-acetylcystein for <i>chronic bronchitis</i>	MOD to HIGH CAT $\geq$ 10, mMRC $\geq$ 2)	HIGH	<80	In addition to LAMA/LABA/ICS : ↓ AECOPD Vs placebo	Weak recommendation, Lower certainty of evidence. Recommend AGAINST if low symptom or Low AECOPD Risk

LOW Risk of AECOPD: $\leq$ 1 moderate exacerbation in last year (rx abx or pred, no ED/hosp)

High Risk of AECOPD:

$\geq$ 2 moderate or $\geq$ 1 severe exacerbation in past year requiring ED visit/hospitalization.

COPD Rx to Prevent Acute Exacerbations

- **Recommended** (= Gr1, **definitely reduces exacerbations**)
 - Annual flu vaccine
 - Pulm Rehab (if **RECENT** exacerbation <4 weeks ago)
 - Education and Case Management
 - Inhaled pharmacotherapy as per previous slides
 - **NEW GOLD 2025:** Dupilumab (see next slide)
- **Suggested** (“consider these” – mixed evidence of reduced exacerbations, but logically they will help COPD patients, Gr2C recommendation CTS 2017)
 - Pneumococcal vaccination
 - *Since CTS 2017 there’s a good Cochrane Review demonstrating NNT=8 for pneumococcal vaccination to prevent AECOPD (and reduces CAP in pts w COPD), so practically recommend to all! (Walters JA et al, 2017)*
 - Smoking Cessation [**this reduces mortality and CV events**. This is exam MCQ fodder, practically ask everyone to stop smoking]
 - Do not use the following to prevent AECOPD: systemic corticosteroids >30d after initial exacerbation, theophylline

Bonus: Read
on your OWN

GOLD 2025: What's new in COPD management

FYI only

- **Dupilumab:** reduces exacerbations and improve FEV1 and QOL (BOREAS 2023 and NOTUS 2024)
 - Rate of moderate or severe COPDe compared to placebo (RR 0.70, 95% CI 0.58-0.86, $p < 0.001$)
 - **GOLD 2025 indications:** Patients on triple inhaled therapy, with symptoms of chronic bronchitis, eos ≥ 300 and ongoing exacerbations
 - Approved by Health Canada as of Oct 2025, not yet covered by ODB
- **Ensifentrine:** novel inhaled PDE3/4 inhibitor with anti inflammatory and bronchodilator effects
 - Phase 3 clinical trials ENHANCE1 & 2 (2023) showed significant decrease in FEV1 decline, dyspnea, and health status compared to placebo
 - Pooled analysis published in Chest in 2025 also found 41% reduction in moderate to severe exacerbation rate (RR 0.59, 95%CI 0.43-0.80, $p < 0.001$)
 - **GOLD 2025 indications:** dyspnea or exacerbations despite LAMA/LABA therapy
 - Not yet approved by Health Canada

Bonus: Read
on your OWN

Therapeutic Options: Systemic Drugs

Drug	Indicated in maintenance treatment of COPD?
Influenza Vaccine	YES, all COPD annually
Pneumococcal Vaccine	YES, all COPD >65, significant comorbid conditions, and all those with FEV1<40%
COVID 19 vaccines	YES, All COPD, all boosters per NACI guidance (CDC, NACI, GOLD...)
TdAP Pertussis vaccine	YES, if not received in adolescence to protect against pertussis
RSV Vaccine	YES, if age >60 and chronic lung disease (GOLD 2024)
Shingrix vaccine	YES, for adults with COPD ≥ 50yrs
Systemic Corticosteroids	NO: Not indicated for maintenance – only use in setting of acute exacerbation
For Patients with MOD-SEVERE COPD: therapies to prevent AECOPD	
Roflumilast (DAXAS) – PDE4 inhibitor	YES – (“ Suggest ”) for patients with chronic bronchitis and high risk of AECOPD despite optimal inhaled therapy
N-Acetylcystine 600mg po BID	YES – (“ Suggest ”) for patients with chronic bronchitis and high risk of AECOPD despite long-acting inhalers
Azithromycin/Macrolides	YES – (“ Recommend ”) for patients high risk of AECOPD despite optimal inhaled therapy
Theophylline	NO – “Recommend AGAINST” use to prevent AECOPD in those on optimal long-acting therapies (Gr2B).

Dyspnea in Advanced COPD (CTS 2011, GOLD 2025)

- **Recommended:**

- oral (but not nebulized) opioids (Grade 2C)
- neuromuscular electrical muscle stimulation (Grade 2B)
- chest wall vibration (Grade 2B)
- walking aids (Grade 2B)
- pursed-lip breathing (Gr 2B)
- continuous oxygen therapy for hypoxemic COPD patients **reduces mortality, and may reduce dyspnea** (Grade 2B)
 - little benefit on QOL

- **Not Recommended:**

- anxiolytic and antidepressant (Grade 2B)
- supplemental oxygen in nonhypoxemic patients

- **Not enough evidence:**

- acupuncture, acupressure, distractive auditory stimuli (music), relaxation, handheld fans, counselling and support programs, or psychotherapy

Asthma+COPD (CTS 2017)

- Some controversy (GOLD: “We no longer refer to asthma-COPD overlap” – called “asthma+COPD” in 2025)
- Asthma+COPD characterized by persistent airflow limitation with several features of both asthma and COPD
 - Worse outcomes than COPD or asthma alone
 - ↑ exacerbation, ↓ QOL, ↓ lung function
 - ↑ mortality
- Suspect ACO when:
 - COPD risk factors and COPD symptoms (chronic bronchitis or chronic dyspnea and exercise limitation rather than episodic)
 - History of allergy or atopy, asthma (childhood, physician diagnosis, wheeze, exacerbation or supportive physiology)
 - Pre and post bronchodilator spirometry (both fixed obstruction post bronchodilator but historic or current reversibility/variability)

Asthma+COPD (ACO)

- **Proposed Diagnostic Criteria (CTS 2017)**

- REQUIRED:

1. Diagnosis of COPD given risk factors, history, spirometry
2. History of asthma (past history/diagnosis, current symptoms consistent, or physiology confirmed /w spirometry)
3. Spirometry: post-bronchodilator fixed FEV1/FVC <0.7

- Supportive but not required:

1. Documentation of a bronchodilator improvement of FEV1 by 200ml or 12%
2. Sputum eosinophils >3%
3. Blood eosinophils >300 cells/uL (current or prev documented)

- **Treatment:**

- Per GOLD: treat as asthma.
- *LABA-ICS combo first line (CTS 2017, CTS 2023) over LAMA-LABA*
- For refractory symptoms, add LAMA to LABA-ICS combo
- Caveat there are no RCTs addressing this population (asthma trials exclude smokers, COPD trials exclude asthmatics)

Advanced Therapies

- NIV in Stable COPD with Hypercapnia
 - **CTS 2022:** Suggest chronic NIV for patients with severe COPD on home oxygen and chronic hypercapnia ($\text{PaCO}_2 \geq 52$)
 - Several trials showing reduction in hospital re-admission rates; **some showing mortality benefit**
- Lung Volume Reduction Surgery:
 - Surgical procedure in which parts of the lung are resected to reduce hyperinflation improving the effectiveness of the respiratory muscles
 - **Increased survival** in severe emphysema patients with upper-lobe predominant disease and low post-rehabilitation exercise capacity (NETT trial 2003)
 - Bullectomy may be considered in select patients with large bulla
- Lung transplant:
 - Bode score 7-10 and 1 of: 1) hospitalized with COPDE with $\text{pCO}_2 > 50$ 2) pulmonary hypertension/cor pulmonale despite supp oxygen 3) $\text{FEV}_1 < 20\%$ with $\text{DLCO} < 20\%$
 - Shown to **improve QOL** and functional capacity
 - Survival benefit in AATD and severely impaired COPD individuals (high BODE)

AECOPD — GOLD 2025

- Acute worsening of respiratory symptoms requiring additional therapy
 - 2023: Prospective severity criteria - proposed, not used in practice
- Most commonly caused by respiratory tract infection
- 1-year mortality after AECOPD is ~30% vs. 23% with MI!
 - Lose additional 8mL/yr of FEV1 with exacerbation
- Treatment:
 - Supplemental oxygen
 - Short-acting BD
 - Long-acting BD initiated ASAP prior to discharge
 - Steroids Pred 40mg x 5 days
 - Shorten recovery and hospitalization duration
 - Antibiotics x 5 d (when indicated)
 - Shorten recovery, ↓ relapse/treatment failure
 - NIV

SEVERITY OF AECOPD	Mild	Worsening symptoms without change in medications
	Moderate	Antibiotic +/- oral steroid
	Severe	Hospitalization/ED visit

GOLD 2025 - Antibiotics

Antibiotics should be given in COPD in the presence of three cardinal symptoms (or two of the following if increased purulence* is one of them) or if patient requires ventilation (NIV or invasive):

1. Increase in dyspnea
2. Increase in sputum volume
3. Increase in sputum purulence

Antibiotics should be given if patients require invasive or non-invasive mechanical ventilation

- Length: 5-7 days [**Recommend $\leq 5d$ if outpatient** based upon meta-analysis]
- Choice of antibiotic based on local resistance pattern. Usual choice includes amox-clav, macrolide, tetracycline or in selected pts, a respiratory fluoroquinolone.

**Increased purulence has ~95% sensitivity and 52% specificity for high bacterial load!*

Do not use
procalcitonin to guide
decision re Antibiotics
(GOLD 2024)

AECOPD — steroids, HFNC and PPV

✓ Steroids

- Benefit: ↓ length of stay. Potentially faster recovery time and increased FEV1
- Prednisone oral equiv 40mg/day (5 days)
- 5 days non-inferior to 14 days (REDUCE trial 2013)
 - **Cochrane meta-analysis 2018** showed no difference between steroid duration <7 or >7days
- May be less efficacious in patients w low blood Eosinophils (GOLD 2024)

✓ BiPAP/NIV

- AECOPD: Preferred over invasive ventilation if no contraindication
- Significant **mortality benefit** and reduction in intubation rate (GOLD 2024)
- Recommended (GOLD 2025) if any of:
 - **pH ≤7.35 with paCO₂ ≥ 45**
 - **severe dyspnea** (impending respiratory failure)
 - **persistent hypoxemia despite supplemental oxygen**

X HFNC – high flow nasal cannula – *not currently recommended*

- Does not prevent intubation in RCT of AECOPD, meta-analysis of poor quality data showed no benefit (GOLD 2023)
- ERS currently recommends trial NIV before HFNC in patients with COPD and acute hypercapnic respiratory failure

Smoking Cessation

ATS 2020: Pharmacologic Treatment in Tobacco-Dependent Adults Only 3% of smokers who are untreated will achieve abstinence within a given year

- Varenicline > nicotine replacement, e-cigarettes, or bupropion
- Varenicline + nicotine patch > varenicline (conditional recommendation, low certainty)
 - Clinicians should begin treatment with varenicline rather than waiting until patients are ready to stop tobacco use (Strong recommendation, moderate certainty)
 - Recommend extended-duration therapy (>12 wk) over standard duration (6–12 wk) therapy

New: CTFPHC 2025 Smoking cessation

- Combined behavioural + pharmacotherapy > either alone
- Acceptable pharmacologic options (no hierarchy given): varenicline, cytisine, bupropion, NRT
- Conditionally recommend against e-cigarettes – consider if failed above
- Strong recommendation against: St. John's wort, hypnotherapy, acupuncture, laser therapy, electrostimulation, continuous auricular stimulation

MCQ 2 2026

A 52-year-old man with depression and schizophrenia who has been smoking 1-2 packs per day since he was 12 years old presents to your clinic with a new pulmonary nodule. At the end of your appointment, you speak with him about smoking cessation however, he is reluctant because he does not think he will be successful. He tried nicotine patches 5 years ago and relapsed quickly. Which of the following is the next evidence-based step?

- A. Counsel patient on the risks of ongoing smoking and revisit smoking cessation at each follow up visit
- B. Start varenicline monotherapy.**
- C. Start bupropion monotherapy.
- D. Start a non-nicotine containing e-cig

Patients started on pharmacotherapy have higher rates of abstinence even when they are not ready to quit . There used to be a black box warning regarding neuropsychiatric adverse effects for varenicline due to case reports in post marketing surveillance. The EAGLES RCT found no increased neuropsych AE with varenicline compared to placebo, even in patients with psychiatric disease.

Take a minute to Catch your breath...



Postop Shortness of Breath...
(Source- Dr McKenna, used w patient permission)

Pleural Effusion

- Go to guidelines/resources:
 - **British Thoracic Society Pleural Disease Guideline 2023**
 - Statement on the management of pleural infection in adults. ERJ 2023
 - Management of Malignant Pleural Effusions. An Official ATS/STS/STR Clinical Practice Guideline. (2018)
 - Evaluation of the patient with pleural effusion. CMAJ March 2018



MCQ 3 (2026)

A 73-year-old retired plumber presents with 6-week history of progressive orthopnea and exertional intolerance. CT Chest shows a small right sided pleural effusion. There are also bilateral calcified pleural plaques without pleural thickening or nodularity. A diagnostic thoracentesis is performed. The aspirated pleural fluid is serosanguinous. Cell count and differential shows 15% neutrophils, 23% lymphocytes, 43% eosinophils, and 19% macrophages. Pleural studies: pH 7.35 LDH 280 protein 30. Serum studies: LDH 320 protein 46. Culture and cytology are both negative. What is the most likely diagnosis?

- a) Mesothelioma
- b) Complicated parapneumonic effusion
- c) Benign asbestos related pleural effusion
- d) Tuberculosis

Pleural Effusion (BTS/Chest)

- When to consider a thoracentesis:

- Suspect exudate
- Unclear cause
- Parapneumonic effusion (**sample fluid if >1 cm on lateral decubitus CXR** – from old CHEST 2000 guidelines prior to widespread availability of US. Not mentioned in BTS 2023 but may see old questions)

– **Image guidance (US) should be used** to reduce risk of complications (BTS 2023)

Aspiration should **NOT** be performed for bilateral effusions in a scenario strongly suggestive of a transudate/heart failure.

Exudate NYD:

- Can repeat thoracentesis **once** for increased sensitivity; sensitivity of pleural fluid cytology for malignancy increases to approx 80% with two thoracenteses
 - At least 25cc, preferably 50cc
- Get CT chest
- Refer for pleural biopsy:
 - Can do CT guided if area of obvious abnormality
 - Thoracoscopy with pleural biopsy is preferred test
 - Mesothelioma usually **cannot** be diagnosed from pleural fluid and requires pleural biopsy

Light's Criteria

Pleural fluid is an exudate if any ≥ 1 of the following criteria are met:

- Pleural fluid Protein : serum Protein > 0.5
- Pleural fluid LDH : serum LDH > 0.6
- Pleural fluid LDH $> 2/3$ of the upper limit of normal value for serum LDH.

Suspect Malignant Effusion?
sensitivity of pleural fluid for
cytology only 46%!

Pleural Fluid Differentials



- Chylothorax – (TGs >1.24mmol, +Chylomicrons)
 - Malignancy #1 (**most commonly lymphoma**)
 - Trauma/surgery, TB, LAM (lymphangioleiomyomatosis = young woman with cystic lung disease/pneumothorax, renal angiomyolipomas)
- Pleural fluid eosinophilia - (>10%)
 - Etiology depends on history
 - Asbestos related (benign asbestos related pleural effusion), drugs (ex. nitrofurantoin), malignancy (lung), infection (parasites), PE, eGPA
- Low glucose - (gluc<3, eff/serum<0.5)
 - Severity: <1 mmol – **RA, empyema**
 - 1-3mmol – malignancy, TB, SLE
- Lymphocytosis
 - >80% TB vs lymphoma
 - If suspecting TB, pleural fluid should be sent for AFB smear/culture and NAAT along with sputum AFB. If non-diagnostic then pleural biopsy should be performed (Canadian TB standards 8th edition)
 - Carcinoma, yellow nail, sarcoidosis, RA

Management of Parapneumonic Effusion (BTS)

- **Chest tube (14F or smaller) indicated if:**
 - drainage of frank pus
 - $\text{pH} \leq 7.2$ [high risk of complicated parapneumonic effusion]
 - $\text{pH} 7.2\text{--}7.4$ and $\text{LDH} > 900$ [consider based on clinical scenario, $\text{glucose} \leq 4.0$, pleural enhancement]
 - **If pH not available, glucose < 3.3**
 - Rule out other potential causes of low pH. I.e. rheumatoid, TB
- **Antibiotics for pleural infection (ERS pleural infections 2023)**
 - Empiric coverage for community acquired - 2nd or 3rd generation cephalosporin with anaerobic coverage (e.g. ceftriaxone and metronidazole)
 - Empiric coverage of hospital acquired pleural infection - covers gram negative and MRSA (e.g. piperacillin-tazobactam + vancomycin)
 - Typically initial 5-7 days of IV Abx followed by prolonged oral course, treat based on culture
- **Usually prolonged antibiotic course (often **at least 3 weeks**, based on clinical and radiographic response)**
 - “2-6 weeks” per BTS, no RCTs to guide length
- **Further treatments like intrapleural lytics and VATS require consultation with Resp or Thoracic Surgeon**
 - Combined tissue plasminogen activator and DNase should be used for treatment of pleural infection where chest tube drainage has ceased and residual collection remains

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- a) Mesothelioma
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- c) Benign asbestos related pleural effusion
- d) Tuberculosis

Calcified pleural plaques in a retired construction worker raises alarm for prior asbestos exposure. BAPE is favoured here over mesothelioma given lack of pleural nodularity/thickening, although early pleural biopsy is still recommended in BAPE to rule out mesothelioma. Pleural cytology is often negative in mesothelioma and cannot be relied on. Both B and D are unlikely due to lack of consistent symptoms/parenchymal changes although isolated TB pleural infections can occur. Typically, B is neutrophilic. D is neutrophilic initially then lymphocytic in late stages, but all listed options can be eosinophilic.

Bronchiectasis

- Good to have very basic approach to diagnosis for the exam

Guidelines for reference: ERS 2017, BTS 2019

- “Bronchiectasis is a chronic respiratory disease characterized by a clinical syndrome of **cough, sputum production and bronchial infection**, and radiologically by abnormal and permanent **dilatation of the bronchi**.
- Most common symptoms:
 - Cough with sputum and/or hemoptysis
 - Dyspnea and fatigue
 - Rhinosinusitis
 - Thoracic pain
- Treatment goals (European Resp Soc Guideline 2017):
 - Prevent Exacerbations
 - Improve QOL
 - ↓ disease progression

STUDY TIP:

Workup & Manage Bronchiectasis is a great scenario to practice with your study group!

Check out our **2015 Physical Exam Video** Online for the exam for “causes and complications of bronchiectasis”

Work up should be directed based on history and physical. Everyone should get chest CT and PFTs (expect obstruction). Most common causes are **post-infectious and idiopathic (accounting for 50% of cases)**

Cause	Clinical Features	Testing
Postinfectious	History of previous infection (i.e., pneumonia, pertussis, NTM, TB)	
Idiopathic	Exclude other causes	
Humoral immunodeficiency	Recurrent infections	Serum immunoglobulins, responses to pneumococcal, H.flu and tetanus antigens
Cystic Fibrosis	Younger age (<45 yrs), history malabsorption, pseudomonas infection, s. aureus, NTM, male infertility.	Sweat chloride, CFTR genetic testing
Autoimmune/CTD	Clinical signs of CTD/vasculitis	RF, anti-CCP, others dependent on history.
Inflammatory bowel disease	Clinical signs of UC or Crohn's	Specialist GI review
ABPA	Hx of difficult to control asthma or CF, blood eosinophils >500, positive aspergillus IgG or positive precipitins, sputum culture of A. fumigatus, mucus plugging/proximal bronchiectasis or fleeting infiltrates on chest imaging	Total IgE >500 IU/mL, positive A. fumigatus specific IgE or immediate reaction on skin-prick testing
Aspiration	History of reflux/aspiration	Video-fluoroscopic swallow study, upper endoscopy, esophageal manometry
Congenital	Williams-Campbell syndrome, Mournier-Kuhn, and lung sequestration	
PCD	History of chronic respiratory tract problems, otitis media, male infertility.	Measurement of nasal nitric oxide levels and ciliated epithelial biopsy.
Alpha-1-AT	Evidence of emphysema/obstruction on spirometry. Panniculitis.	Serum alpha 1 AT level

Bronchiectasis Work-up (BTS 2019)

Routine work up for all:

- Baseline CXR and CT Chest
- Sputum C&S and AFB for NTM (induced/bronch if needed)
- ABPA screening:
 - CBC (for eosinophilia)
 - Total IgE
 - Sensitization to aspergillus (IgE-specific antibodies or skin prick)
- Immunodeficiency screening: serum immunoglobulins (IgG, IgA, IgM) +/- pneumococcal vaccine response with IgG

Consider if supportive clinical features:

- CF: sweat chloride test
- Primary Ciliary Dyskinesia: nasal nitric oxide
- RF, anti-CCP, ANA, ANCA
- Alpha1 anti-trypsin level
- Videofluoroscopic swallow study
- HIV testing

Bronchiectasis Treatment (BTS 2019)

Mainstays of treatment:

- Airway clearance: Active cycle of breathing technique
- Mucoactive agents: Hypertonic saline (**avoid DNase unless CF!**)
- Antimicrobials
 - Bronchodilator: Inhaled colistin or gentamicin (if Pseudomonas (PsA) colonized)
 - Chronic azithromycin if recurrent exacerbations (with or without PsA colonization but rule out NTM)
- Unless otherwise indicated, may offer as trial if significant breathlessness
- Pulmonary rehab: If functionally limited by dyspnea (mMRC ≥ 1)
- Vaccines: Influenza, Pneumococcal, (+ TdAP, COVID, etc per NACI guidelines)
- Supplemental oxygen (same criteria as for COPD)
- **Do not routinely offer ICS, oral steroid, PDE4-I**

NEW: Brensocatib (ASPEN 2025) – **First disease modifying therapy in bronchiectasis** – oral DPP-1 inhibitor blocking neutrophilic inflammation. Significant decrease in bronchiectasis exacerbations vs. placebo (RR 0.79, 95% CI 0.68-0.92) and reduces FEV1 decline by 50%. ****Not Yet available in Canada in 2025****

Advanced Therapies:

- Consider surgery / lung resection in localized disease
- Consider transplant if poor lung function and one of: massive hemoptysis, severe PH, ICU admissions or respiratory failure (require NIV)
- Consider NIV if respiratory failure with hypercapnia, especially if recurrent hospitalization

Bronchiectasis Exacerbation (BTS 2019)

- Obtain sputum cultures
- Empiric antimicrobials, covering organisms previously grown, if known*
 - *no evidence to show antibiotics guided by sensitivity results improves outcomes
- Generally 14 day course recommended, especially if PsA colonized
 - Shorter course possible if mild bronchiectasis
- Consider IV antimicrobials if unwell/admitted, resistant organisms or failed oral therapy
- Major Hemoptysis
 - IV antimicrobials
 - Tranexamic acid
 - Consider embolization as first line



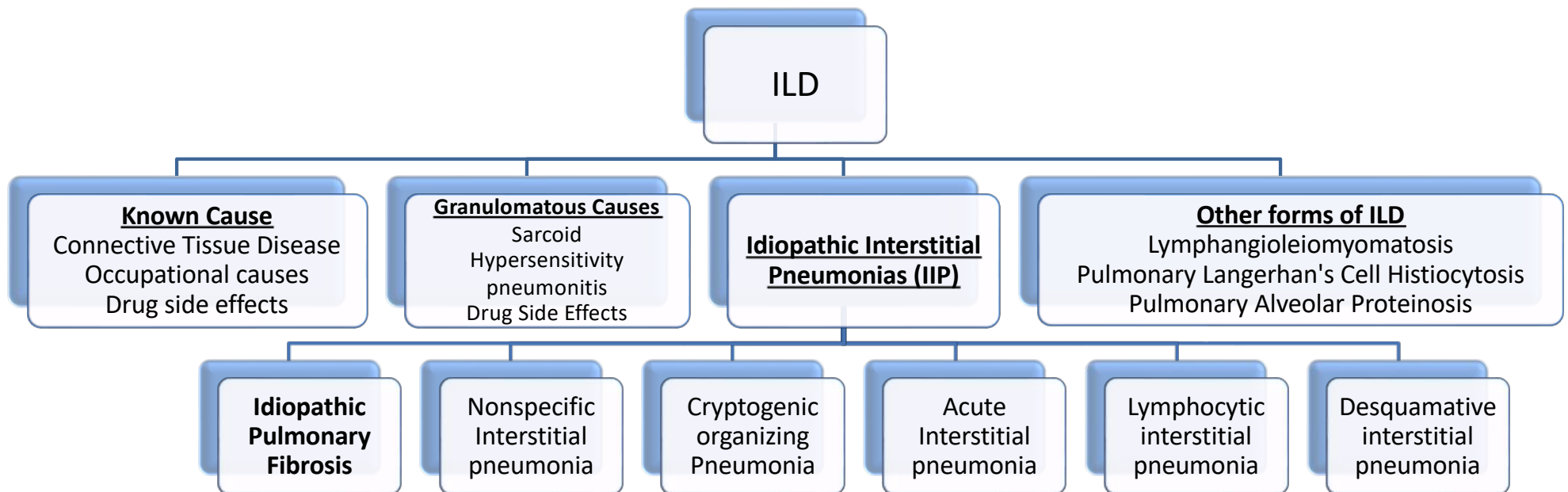
Another good
scenario to practice
managing with
your study group!

Interstitial Lung Disease

- Heterogenous group of diseases (>200 subtypes)
- Go to resources for RC:
 - *Evaluation of patients with fibrotic interstitial lung disease: A Canadian Thoracic Society position statement (2017)
 - IPF and Progressive Pulm Fibrosis in Adults: Official ATS/ERS/JRS/ALAT Clinical Practice Guideline (2022)
 - Diagnosis of Idiopathic Pulmonary Fibrosis: Official ATS/ERS/JRS/ALAT Clinical Practice Guideline (2018)



ILD: Associated with a known disease or not?



Don't worry about these for the purposes of the exam. Just highlighting that there are many different IIPs!

Interstitial Lung Disease- Approach

Differential

- Idiopathic (see next page)
- Familial (rare)
- Drug related (MTX, macrobid, amio, bleomycin, many others)
- Radiation
- Exposure related (immune mediated=hypersensitivity pneumonitis, non-immune ILD=pneumoconiosis)
- SARD related (SSc, RA, IIM, Sjogren's, MCTD)
- Miscellaneous: Sarcoid, Organizing pneumonia, others

History/Physical

- **Family History**
- **Drugs and radiation**
- **Inorganic exposures**
 - asbestos, mining, sandblasting, etc
- **Organic exposures**
 - mould/water damage, birds, farms, feather pillows, hot tubs, etc
- **Connective tissue disease**
 - joints, mechanic's hands, Gottren's, Raynaud's, Shawl sign, heliotrope, scleroderma findings, proximal muscle weakness, etc

Routine work up for all:

- ANA, RF, anti-CCP
- HRCT Chest
- PFTs
- Multidisciplinary Discussion (MDD)

As clinically indicated:

- Specific autoantibodies (myositis) +/- Rheumatology referral
- Precipitating antibodies for some known antigens for HP
- 6MWT/home O2 assessment
- Lung biopsy/BAL

IPF (ATS 2022 Guideline)

- Most common IIP
- M > F, 6th or 7th decade (rare <50y)
- Many have smoking history
- Chronic, progressive, fibrotic ILD defined by the histologic and/or radiographic pattern of Usual Interstitial Pneumonia (UIP)

Diagnosis requires:

1. Exclusion of known causes of ILD
2. UIP pattern on high-resolution CT (see next page)
3. **Consensus through multidisciplinary discussion (MDD)**

Radiology (ATS 2022)

UIP Pattern

1. Reticular changes
2. Subpleural/Basal predominant
3. Honeycombing
4. Absence of inconsistent features →

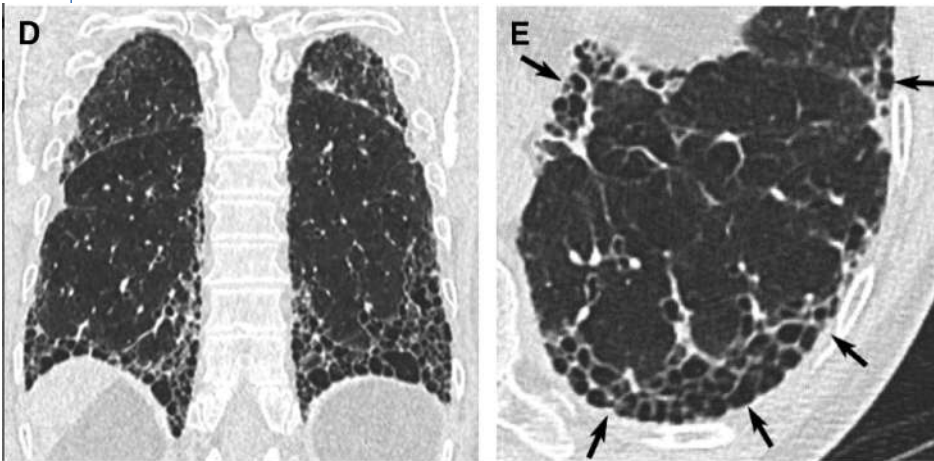
Alternative Diagnosis

Features:

- Cysts
- Mosaic attenuation
- Predominant GGO
- Profuse centrilobular micronodules
- Nodules
- Consolidation

Predominant Distribution:

- Peribronchovascular
- Perilymphatic
- Upper or mid-lung



IPF Treatment

Treatment:

- Quit smoking, pulmonary rehab, Pneumovax and influenza vaccination (+COVID, + update Tdap per NACI...) +/- home O2, transplant/palliative referral
- Anti-fibrotic medications (both reduce decline in FVC by about 50%):
 - Choice based on pt preference, side effect profile
 - **Nintedanib** (INPULSIS 1/2 trials) – TK inhibitor, resulted in reduced FVC decline, pooled analysis shows *trend* to reduced mortality. Adverse effects: diarrhea, GI upset, transaminitis
 - **Pirfenidone** (ASCEND/CAPACITY trials) – reduced FVC decline, 6MWT decline, pooled analysis showed improved survival. Adverse effects: photosensitivity, GI upset, transaminitis
 - **NEW: Nerandomilast** (FIBRONEER-IPF 2025) – **An add-on anti fibrotic?** 80% of study participants already on nintedanib/pirfenidone. Add on nerandomilast reduced FVC decline by a further 40%!
- **No role for chronic corticosteroids or immunosuppression given ↑mortality (PANTHER-IPF)!**
- Routine PPI not indicated to improve resp outcomes (Rx if indicated for GI reasons) –ATS 2022

Acute exacerbation:

- Worsening dyspnea, hypoxemia with new diffuse bilateral ground glass/consolidation on CT
- Rule out infection/PE/heart failure
- Consider high dose corticosteroids (no evidence) and empiric antimicrobials
- Dismal prognosis – *generally 50% in-hospital mortality*

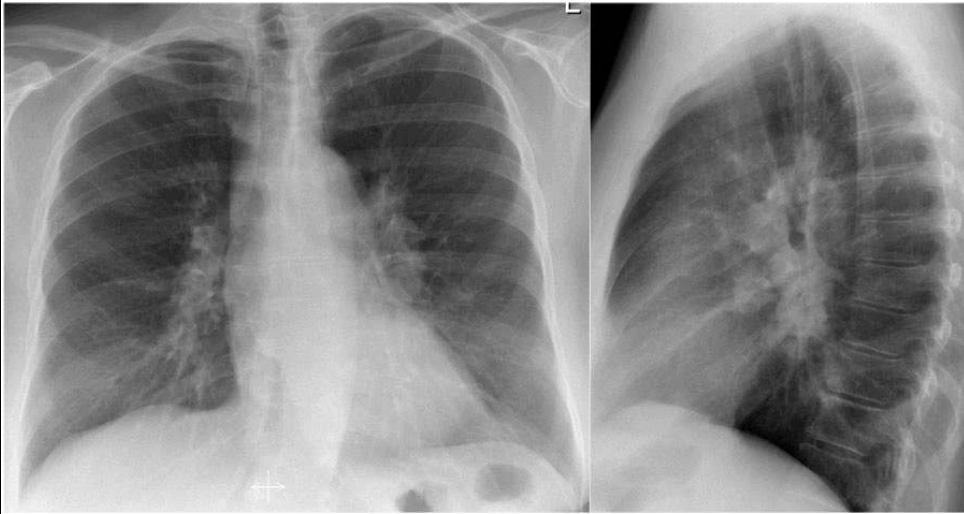
CTS 2018 ILD Position Statement Summary

- Accurate diagnosis is key
- Rx IPF w/ anti-fibrotics (pirfenidone, nintedanib), consider immunosuppression in non-IPF
- Pulmonary rehab is recommended
 - ↓ dyspnea, ↑ walking distance and QOL
- There is a role for supplemental oxygen in:
 - Resting hypoxemia – Same criteria as COPD
 - Exertional hypoxemia (sat <88%) **with** improved walk distance or dyspnea on supplemental oxygen
- Advanced care planning should be addressed in all patients with fibrotic ILD in course of their disease (+ early palliative care referral for symptom control)
- Dyspnea management
 - Non-pharmacologic techniques and low-dose opioid (morphine 1.0-2.5mg q4-6h)
- Cough management
 - Opioids may be considered first-line in refractory cases
- Consider lung transplant referral early for IPF and fibrotic NSIP (for Resp consultant!)
 - Criteria for referral: FVC <80%, DLCO <40%, need oxygen, 'failed' pharmacotherapy

Non-IPF ILD

Class	Cause	Imaging (FYI only)	Treatment
Hypersensitivity pneumonitis	Immune mediated hypersensitivity reaction to inhaled antigens (moulds, birds, chemicals, drugs, etc.)	New 2025: "Bronchocentriolar pneumonia" (BIP) = 'UIP' equivalent of HP Upper lobe predominant GGOs, nodules, 3-density sign +/- fibrosis	Avoid antigen Steroids MMF or AZA
Pneumoconiosis	Retained inhaled dusts (often occupational: asbestos, silica, etc)	Variety of imaging patterns	Avoid exposure (remove from work, PPE) Worker's compensation
SARD-ILD *Previously 'CTD-ILD' (terminology updated in ACR 2023)	Scleroderma, RA, IIM, Sjogren's, Mixed Con Tissue Disease	Variety – UIP, NSIP, BIP, organizing pneumonia, and others	Immunosuppression – see Rheumatology section on SARD-ILD (*avoid steroids in SSc - renal crisis risk)
Drug-induced ILD	Methotrexate, amiodarone, nitrofurantoin, bleomycin, vaping	Variety – HP, NSIP, BIP, organizing pneumonia, and others	Stop the drug Steroids

Sarcoid



Guideline

ATS 2020 Guidelines on the Diagnosis and Detection of Sarcoidosis

ERS 2021 Clinical Practice Guidelines on Treatment of Sarcoidosis

- Multisystem **granulomatous** disease of unknown etiology
 - Pathology reveals noncaseating granulomas in involved organs
 - Usually involves the lungs (>90%) but up to 30% of patients present with extrapulmonary sarcoid including:
 - **Heart, CNS, eyes*** need urgent treatment!
 - Other: liver, spleen, musculoskeletal system, kidney
- Most are asymptomatic
 - 2/3 of patients will have remission at a decade
 - After one year of spontaneous remission relapse is very uncommon

****HIGH YIELD****

Diagnosing sarcoidosis

ATS 2020 diagnostic criteria (highlights these are not definitive thus “diagnosis of sarcoidosis is never fully secure”):

1. Compatible clinical presentation
2. Non necrotizing granulomatous inflammation (i.e. need biopsy for diagnosis **except** in below syndromes)
3. Exclusion of alternative granulomatous causes

- Lofgren Syndrome
 - **Bilateral hilar adenopathy**
 - **Erythema nodosum**
 - **Migratory polyarthralgias**
 - Fever
 - Seen primarily in women
 - High likelihood of spontaneous remission
- Heerfordt Syndrome
 - **Anterior uveitis**
 - **Parotid enlargement**
 - **Fever (uveoparotid fever)**
 - Facial palsy

ATS 2020 recommends:

1. Lofgren syndrome, Heerfordt syndrome, lupus pernio: No biopsy required for diagnosis.
(*Conditional recommendation, very low quality of evidence*)
2. Asymptomatic bilateral hilar adenopathy: no recommendations for/against. Close clinical follow up required if no biopsy.
3. EBUS guided lymph node sampling, rather than mediastinoscopy, is recommended as the sampling method of choice (*Conditional recommendation, low quality of evidence*)

Sarcoid: Pulmonary Involvement

- 90% have pulmonary involvement
 - often asymptomatic hilar lymphadenopathy
- Stage reflects chance of spontaneous remission
- Symptoms can vary: dyspnea, cough, chest pain, wheeze
- *PFTs can show anything*: normal, restriction, obstruction, or both (+/- reduced DLCO)
- Co-existent asthma is common
- Pulmonary hypertension is rare but described

Stage	Rate of spontaneous remission	Chest X Ray Findings
1	55-90% [75%]	Bilateral adenopathy
2	40-70% [50%]	Adenopathy + parenchymal lesions
3	10-30% [25%]	Parenchymal lesions with no adenopathy
4	0-5% [0%]	Fibrosis

Sarcoidosis: Extrapulmonary Involvement

- **Cutaneous:** Common (about 1/3 of patients), lupus pernio, erythema nodosum, others
- **Liver and Spleen:** ~10% of patients, elevated liver enzymes, cholestasis, rarely liver failure, can see liver and spleen lesions on imaging
- **Neurologic:** Cranial nerve palsy, optic neuritis, headache, ataxia, weakness, LP nonspecific lymphocytic inflammation, MRI imaging test of choice
- **Ocular:** Anterior uveitis most common
- **Cardiac:** ~5% of patients clinically (more at autopsy), cardiomyopathy, arrhythmia, heart block, screen with ECG +/- echo, then cardiac MRI and PET scan if concern
- **Hypercalcemia:** Common, rule out other causes of hypercalcemia, due to increased conversion of 25-OHD3 to 1,25-OH₂D3.



Lupus Pernio
nose/cheeks/lips/ ears/digits
Source: Dermnetnz.org

Erythema Nodosum
Panniculitis usually anterior shins
Source: Dr McKenna office



Investigations

- **CXR +/- CT, PFTs**
- **Tissue biopsy: LN or transbronchial biopsy**
- CBC may show anemia of chronic disease, **lymphopenia**, eosinophilia thrombocytopenia
- **Calcium**, Alb, LFTs including ALP to screen for hepatic sarcoid, Cr
- **ECG**
 - If cardiac disease **suspected** – Cardiac MRI is preferred, or cardiac PET if unavailable
 - If Pulm HTN **suspected** – do TTE
- **1,25- OH₂ Vitamin D**
- **Ophtho referral to r/o sarcoid ocular**
Rule out: HIV serology, Rule out TB
- Bronchoscopy/BAL: low CD8, elevated CD4/CD8 ratio
- Serum ACE level is not necessary
 - increased ACE is ~70% sensitive and 90% specific with active disease;
 - Reflects granulomatous load and can be false positive in other granulomatous disease

Sarcoidosis treatment

General principles of treatment

- Most patients with pulmonary sarcoidosis do **not** require treatment – rate of remission is generally high and recurrence is low. They do still require ongoing monitoring with PFT/imaging over subsequent years.
- **Steroid treatment is long (very slow taper over ~12 months) and increases risk of disease relapse: not to be taken lightly. Avoid treatment unless truly indicated.**

Indications to treat with steroids

1. **Extrapulmonary:** eyes, heart, CNS, hypercalcemia, refractory skin disease
2. **Pulmonary:** significant symptoms and progressive lung function decline on PFTs (“high risk of mortality or permanent disability”)
 - If progressive fibrosis (stage 4), often requires functional testing such as a PET scan to assess for underlying inflammation
 - *FYI only - not in guidelines: for milder pulmonary disease (cough, dyspnea), can try ICS or methotrexate (PREDMETH trial 2025)*

When to treat with alternatives

1. Lofgren’s syndrome/erythema nodosum: NSAIDs
2. Other cutaneous sarcoidosis: topical steroids
3. Fatigue: pulmonary rehab
4. Many steroid sparing agents if steroids started: MTX, AZA, mycophenolate, HCQ, TNFa-i (e.g., infliximab), etc.

MCQ 4 2026

An 18-year-old male scuba diver visiting Canada from Florida presents to the ED with sudden onset left sided chest pain. CXR shows a large left apical bulla and 4.2 cm lateral pneumothorax. He is admitted by Thoracics surgery and undergoes a bullectomy and chest tube insertion. On day 6 of admission his lungs expand fully and the chest tube is removed. A repeat chest X-ray 1 hour after chest tube removal shows no recurrent pneumothorax. He is ready for discharge. Which of the following is most true:

- a) He should avoid scuba diving for the next 7 days.
- b) He should avoid physical activity for the next 7 days.
- c) He should avoid flights for the next 7 days.
- d) He should avoid flights for the next 14 days.

Primary Spontaneous Pneumothorax

- Primary: Absence of underlying lung disease
- Risk factors = **smoking***, family history, Marfan syndrome, thoracic endometriosis
 - Reoccur 25-50%, most in first year
- **Conservative management** for asymptomatic or minimally symptomatic patients (no significant pain, breathlessness or physiological compromise)
 - **REGARDLESS OF SIZE OF Pneumothorax (BTS 2023 ... Beware old MCQs refer to 2cm cutoff)**
 - **Conditional recommendation – conditional on reliable patient, close followup plan etc.**
 - **Ambulatory management could be considered**
- If symptomatic, look for **high risk features**: (tension, significant hypoxia, bilateral, >50 with smoking history, hemopneumothorax).
 - If present: admit and insert chest drain
- For all other patients: based on patient preference and local resources
 - Conservative care, ambulatory device, needle aspiration OR chest drain.
- Close follow-up required for conservative care or ambulatory device (every 2-4 days). Follow-up 2-4 weeks for all

Other – recommend against scuba “lifelong”,
no Air travel until 7d after PTx resolves on CXR

Updated: BTS 2023

Secondary Spontaneous Pneumothorax (SSP):

Secondary: *presence of lung disease*

- COPD>>> Others (CF, TB, PJP)
- Even small SSP can lead to significant symptoms; often requires admission
- Most likely will require chest drain given underlying lung abnormalities and risk of air leak. Unlikely to spontaneously resolve.
- Consult Resp/Thoracic Surgery

MCQ 4 (2026)

An 18-year-old male scuba diver visiting Canada from Florida presents to the ED with sudden onset left sided chest pain. CXR shows a large left apical bulla and 4.2 cm lateral pneumothorax. He is admitted by Thoracics surgery and undergoes a bullectomy and chest tube insertion. On day 6 of admission his lungs expand fully and the chest tube is removed. A repeat chest X-ray 1 hour after chest tube removal shows no recurrent pneumothorax. He is ready for discharge. Which of the following is true:

- a) He should avoid scuba diving for t
- b) He should avoid heavy lifting for t
- c) He should avoid flights for the nex
- d) He should avoid flights for the nex

BTS 2023: He should be advised to avoid air travel for 7 days following radiographic resolution of a pneumothorax (to monitor for early recurrence). He unfortunately needs to avoid scuba diving lifelong unless he has definite treatment like surgical pleurectomy. ERS 2024 reviews that while there are case reports linking activity/Valsalva maneuvers to pneumothorax, there is not strong enough evidence to provide recommendation on activity overall post-PTX.

Pulmonary Hypertension (PH)

Relevant guidelines: CCS/CTS 2020 Position Statement on Pulmonary Hypertension, ESC/ERS 2022 Guidelines for the diagnosis and treatment of Pulmonary Hypertension

NEW: 2024 7th World Symposium on Pulmonary Hypertension

- Defined as: **mean pulmonary artery pressure >20mmHg** on right heart catheterization
- Hemodynamic subtypes dependent on PVR (cut off >2WU for pre-capillary, previously >3WU) and wedge
 - $PVR = \text{Pulmonary Vascular Resistance} ((mPAP - PCW) / Q)$
 - PCW = Pulm Cap wedge P and Q = Cardiac output
 - Hx: slowly progressive dyspnea on exertion and eventually right ventricular failure
- Suspect in dyspnea on exertion, PFT with **isolated reduced DLCO**
 - Most important screening is **echocardiogram** – Assess how RV is doing!
 - Probability of PH (low, intermed, high) calculated based upon peak TR velocity, mean PA pressure, other echo features
 - **PH Probability intermed or high → refer to PH Centre and do comprehensive workup**
- Initial work up directed at identifying PH etiology
 - Classify into a **WHO group** (determines management)
- Everyone should get:
 - CBC, lytes Cr LFTs, TSH, NT-proBNP, screening connective tissue disease, viral hepatitis, HIV, ABG
 - Transthoracic echo
 - PFTs for all, +/- 6MWT
 - CT pulmonary angiogram. (+ V/Q scan if unexplained PH to rule out CTEPH)
 - Abdo US to screen for portal HTN
 - Sleep study if suspicious clinical features
 - Right heart catheterization is gold standard

Pulmonary Hypertension – 7th World Symposium 2024

- Patients with a diagnosis of PH should be evaluated/treated at a specialized PH centre!
- Notes from this guideline applicable to GIM:
 - Screen patients with scleroderma annually for PH with echo and DLCO
 - Echo to screen for PH in patients with portal hypertension undergoing transplant w/u
 - If dyspnea or exercise intolerance after at least 3 months of uninterrupted anticoagulation post acute PE assess for CTEPH with echo and V/Q lung scan
 - Patients with CTEPH – test for APLAS

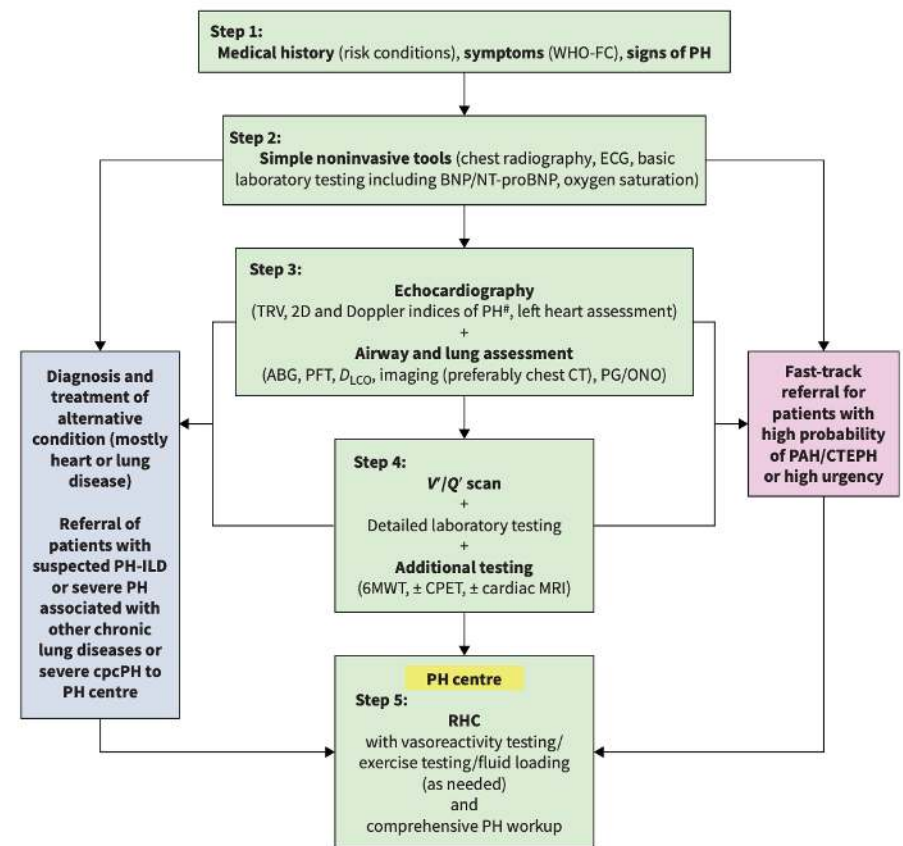


FIGURE 1 Suggested diagnostic approach to pulmonary hypertension (PH). Steps 1–5 represent the most important diagnostic steps of PH from the first presentation of the patient with symptoms or an existing risk condition towards final diagnosis with invasive assessment. WHO-FC: World Health Organization functional class; BNP: brain natriuretic peptide; NT-proBNP: N-terminal pro-BNP; TRV: tricuspid regurgitation velocity; 2D: two-dimensional; ABG: arterial blood gases; PFT: pulmonary function testing; D_{LCO} : diffusion capacity of the lung for carbon monoxide; CT: computed tomography; PG: polygraphy; ONO: overnight oximetry; V/Q scan: ventilation/perfusion scan of the lung; 6MWT: 6-min walk test; CPET: cardiopulmonary exercise testing; MRI: magnetic resonance imaging; RHC: right heart catheterisation; PH-ILD: pulmonary hypertension associated with interstitial lung disease; PAH: pulmonary arterial hypertension; CTEPH: chronic thromboembolic pulmonary hypertension; cpcPH: combined post- and pre-capillary PH. *: refer to figure 2.

For your Reference only! Clues for PH on ECHO:

RVSP (not a hard cutoff/not sensitive) but ≥ 40 mmHg
 “RV enlargement” “RV overload” “IVS flattening”, or TR jet velocity ≥ 2.8 , TAPSE < 1.7 cm, FAC $< 35\%$ – GIM do not need to know ECHO criteria, just when to suspect/refer.

Pulmonary Hypertension (PH)

Group 1: Pulmonary arterial hypertension

- Idiopathic
- Heritable
- Drug/toxin induced
- Associated with: CTD, HIV, PoH, schistosomiasis
- PVOD: pulm venous occlusive disease – rare subtype with occlusion of pulm VEINS

Treatment: Should be referred to a PH centre

- Vasoreactivity testing in I/D/H-PAH for CCB
- **PDE5 inhibitor** or riociguat (sGC inhibitor)
- **Endothelin receptor antagonist**
- **Prostanoids**
- **New! 2024 World Symposium: Sotatercept** (STELLAR 2023, ZENITH 2025) – first therapy to improve major clinical endpoints (reduced composite of death, transplant, hospitalization by 76% vs placebo)
- Supervised exercise, vaccines, consider transplant
- Pregnancy counselling (avoid pregnancy)

Group 2: Due to left heart disease

- Systolic/**diastolic dysfunction**, valvular disease
- Treat underlying cause

Group 3: Due to lung disease/hypoxemia (COPD, ILD, etc)

- Treat underlying cause

Group 4: Chronic thromboembolic pulmonary hypertension

- Lifelong anticoagulation for all
- ***APLAS testing at dx, If + → warfarin* (ERS 2022)**
- Refer for pulmonary thromboendarterectomy

Group 5: PH with unclear and/or multifactorial mechanisms

- Heme disorders (myeloproliferative disease, splenectomy)
- Systemic disorders (sarcoid, LCH, LAM)
- Treat underlying disease

Hypoxemia

Assessing Hypoxemia:

A-a Gradient: P_AO_2 — P_aO_2
estimated from alveolar gas equation measured from ABG

$$A-a = [FiO_2 * (P_{atm} - P_{H_2O}) - (P_aCO_2/R)] - P_aO_2$$

On room air at sea level the first term simplifies to approximately 150

Respiratory Quotient (R) Depends on carbohydrate metabolism but usually 0.8 is used

$$A-a = [150 - (P_aCO_2/0.8)] - P_aO_2$$

Approximate Normal A-a Gradient for reference: $(Age\ (yrs)/4) + 4$

Hypoxemia

Normal A-a gradient (normal gas exchange)

- Hypoventilation
 - CNS depression (drugs, stroke, tumour, bleed, meningitis), spinal cord injury, chest wall abnormality, diaphragm dysfunction (ie phrenic nerve injury or myopathy), neuromuscular disorder, obesity hypoventilation
 - **Check PaCO₂ – if elevated think of hypoventilation**
- Low inspired FiO₂ (e.g. altitude)

Widened A-a gradient

- V/Q mismatch (improves with 100% FiO₂)
 - E.g. COPD, PE
- Shunt (does not improve completely with administration of 100% FiO₂)
 - CAUSE: intracardiac with R->L shunt (eg PFO, ASD, VSD), intrapulmonary (ie pulmonary AVM), physiologic (ie severe pneumonia with perfused alveoli that are not ventilated)
- Diffusion Abnormality
 - E.g. ILD

Hypoxemia

- When to prescribe Home O2
 - PaO2 $\leq$ 55 *or*
 - (Resting SaO2 $\leq$ 88%) *or*
 - PaO2 = 55-59 with:
 - Cor pulmonale *or*
 - Pulmonary hypertension *or*
 - Persistent erythrocytosis (Hct >55%)
- Based trials from the early 1980s showing mortality benefit in COPD, extrapolated to all lung disease
- FYI - Other funded criteria (Ontario - varies by province)
 - Nocturnal hypoxemia [needs sleep study]
 - *Exercise criteria* (beyond scope of this exam*)
 - Palliative setting (90 d funding)

*Long Term Oxygen Therapy Trial (NEJM 2016)
Pts: stable COPD, SpO2 >88% at rest *with exercise induced desaturation*.

Results: No improvement in any measurement – mortality, hospitalization, symptoms, or exercise

Respiratory Blood Gases

RESPIRATORY ACIDOSIS

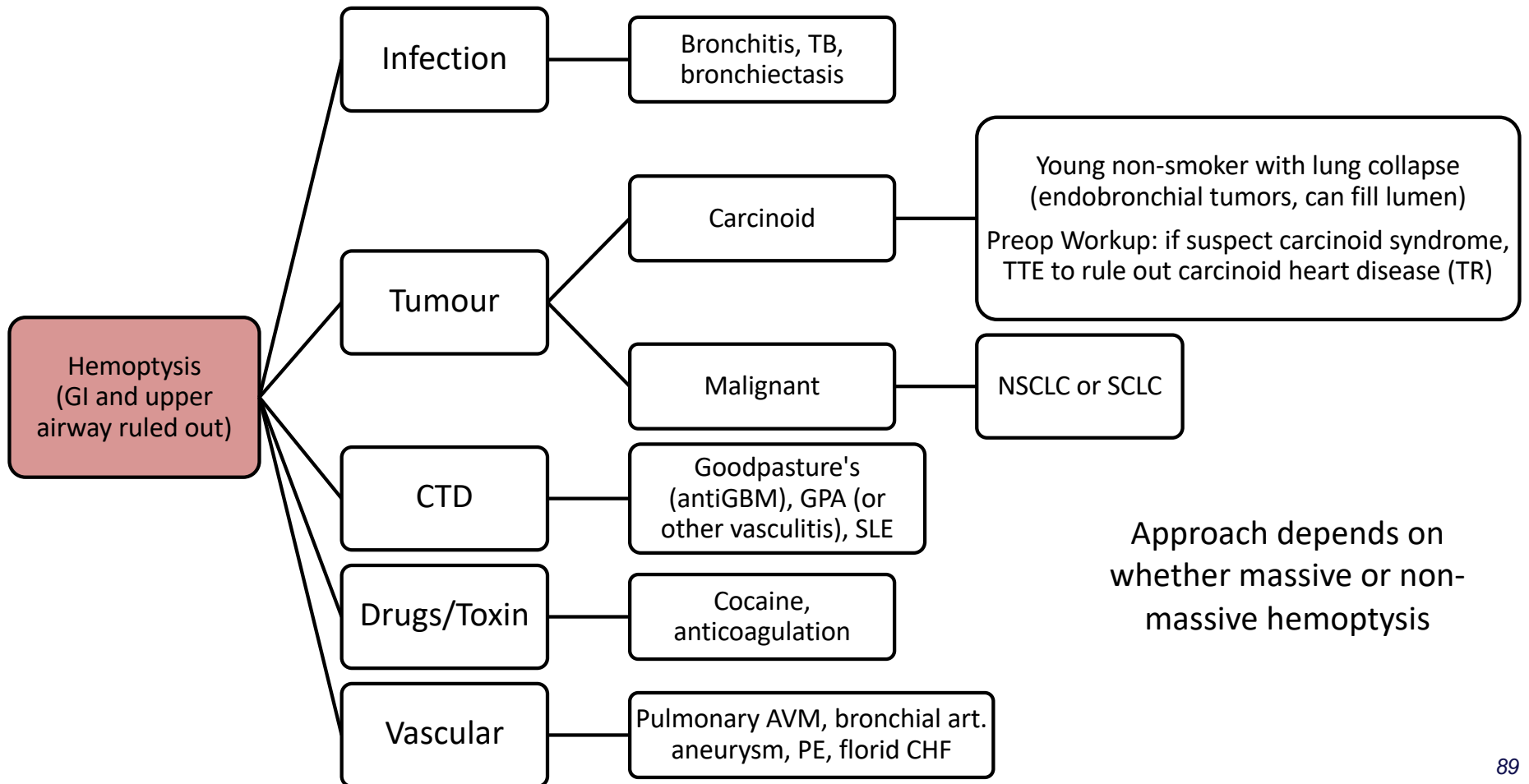
Reflects a decrease in alveolar ventilation. If marked limitation in pulmonary reserve (increased deadspace), or **hypoventilation** will lead to consequent acidosis.

RESPIRATORY ALKALOSIS

Reflects an increase in alveolar ventilation (i.e., pain, fever, anxiety, liver disease, pregnancy) In the presence of mechanical ventilation, decreased carbon dioxide production (i.e., sedation, skeletal muscle paralysis, hypothermia) can cause respiratory alkalosis

Metabolic compensation	RESPIRATORY ACIDOSIS $\Delta p\text{CO}_2 : \Delta \text{HCO}_3$	RESPIRATORY ALKALOSIS $\Delta p\text{CO}_2 : \Delta \text{HCO}_3$
ACUTE	10:1	10:2
CHRONIC	10:3-4	10:4-5
	For every rise of 10 of $p\text{CO}_2$ the HCO_3 will rise by 1 (or 3-4)	For every fall of 10 of $p\text{CO}_2$ the HCO_3 will fall by 2 (or 4-5)

Hemoptysis Differential



Approach depends on whether massive or non-massive hemoptysis

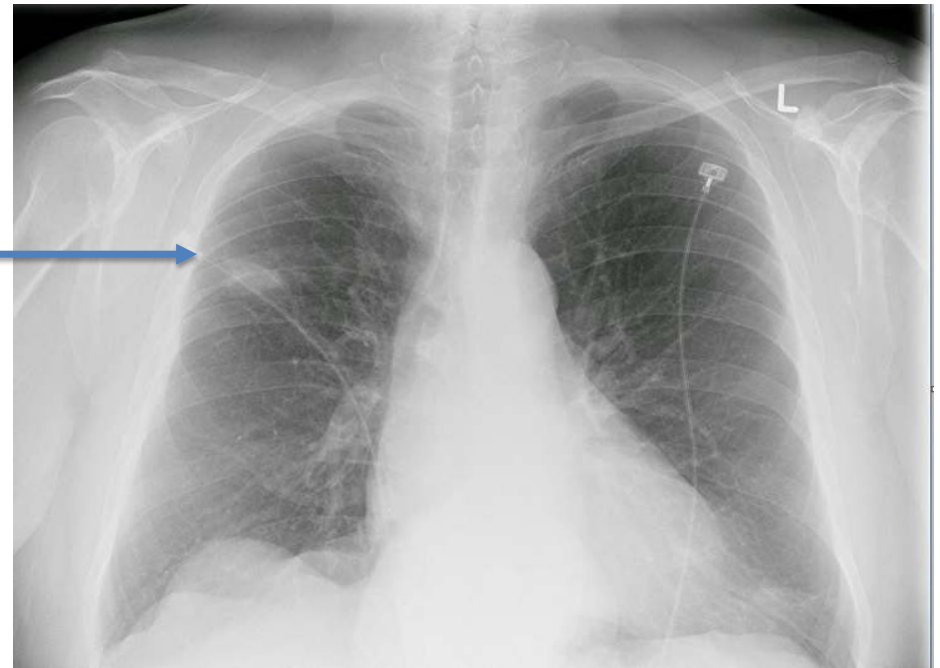
Hemoptysis

- Massive (variable definitions; ~200-600 cc/24 hours)
 - Acute management → Airway, Breathing, Circulation
 - If able to localize bleeding side
 - position patient in decubitus position to protect unaffected lung
 - Hold anti-coagulation/ correct coagulopathy
 - CBC, coagulation profile, ABG
 - Cross-type, IV fluids, oxygen
- Investigations
 - Need imaging if patient is stable enough (CXR/CT)
 - Flexible bronchoscopy can be used for localization prior to IR guided embolization
- Management dependent on local center
 - Interventional Radiology – Arterial embolization #1 if available
 - Thoracic Surgery – Ability for rigid bronchoscopy/Resection
 - **Patients die of asphyxiation NOT exsanguination**

QUESTIONS?

- Be Sure to Check out BONUS Slides in Your Course Pack!
 - Chronic Cough Approach (ACCP)
 - OSA
 - Lung Nodules (Fleischer 2017 guidelines)
 - Smoking Cessation Counselling
 - ** HIGH YIELD for Oral Scenarios**
 - A Couple more slides on less high yield ILD subtypes
 - Relevant Asthma and COPD trials
 - Clubbing
- **PFT Approach, Some PFT “CLANG ASSOCIATIONS” to know **
 - Our ONLINE PFT PRIMER also covers these with examples and chances to PRACTICE your PFT skills!

CXR Approach



Cough (ACCP)



Acute
(<3 weeks)

- Life threatening: Pneumonia, COPDE, asthma, PE, heart failure, other serious disease
- Non-life threatening: Infectious (URTI or LRTI), Environmental / occupational (RADS)

Subacute
(3-8 weeks)

- Post infectious: Pneumonia, bronchitis, pertussis, worsening asthma, GERD, UACS, COPD, NAEB
- Non-infectious (same as chronic ddx)

Chronic
(>8 weeks)

- UACS – upper airway cough syndrome, asthma, NAEB, GERD, smoking, ACEi use

Cough Approach



- CXR and exclude ACE-inhibitors
- In this order, test and treat for:
 - UACS
 - Confirmed by response to Tx with 1st generation antihistamine/ decongestant (bronpheniramine/SR pseudoephedrine)
 - Sinus imaging if does not improve
 - Asthma
 - Methacholine challenge and/or empiric ICS (may treat NAEB as well)
 - GERD
 - If “clinical profile” fits (not necessarily GI symptoms), treat empirically
 - Lifestyle: coffee, chocolate, citrus, smoking ,etc.
 - H2RA/PPI
 - Prokinetic
 - 24h pH monitoring on therapy if cough persists
 - Antireflux surgery if refractory
 - Non asthmatic eosinophilic bronchitis – similar pathology to asthma with airway inflammation but no bronchospasm.
 - Sputum eosinophils, if available, or an empiric trial of inhaled corticosteroids

Obstructive Sleep Apnea

- Defined by presence of both:
 - Symptoms (daytime sleepiness, nocturnal choking/gasping, morning headaches, etc) and
 - Objective testing (≥ 5 apnea/hypopnea events during sleep monitoring)
 - Mild 5-15 (events/h) / Moderate 15-30 / Severe >30
- Treatment Indications
 - Patients with symptoms of excessive sleepiness or impaired sleep-related QOL
 - Comorbid HTN
 - Asymptomatic patients with severe OSA
- Treatments
 - Weight loss (see endocrinology Bonus slides re: Obesity)
 - PAP (CPAP or APAP)
 - All patients should be offered therapy, asymptomatic patients should be treated if have comorbidities (ie. HTN), AHI >30 , critical occupation
 - Oral appliances – for mild-moderate disease
 - Surgery (rare) – Tonsillectomy if appropriate, uvulopalatopharyngoplasty in select patients
- **Evidence:** CPAP for Prevention of Cardiovascular Events in Obstructive Sleep Apnea, NEJM 2016 (RCT). Use of CPAP DID NOT reduce cardiovascular events for asymptomatic patients with moderate-severe OSA. Significance of this result still heavily debated.

Go to resource: 2011 CTS Guideline on Sleep Disordered Breathing, JAMA RCE for OSA, 2019 AASM Guideline on OSA and PAP

Driving and Sleep Disorders

CMA 2023 Driving Guidelines

- Patients with OSA have more collisions than those without as a group, but severity of OSA alone is not a reliable predictor of collision risk and should not be used in isolation to assess fitness to drive
 - Consider co-morbidities, meds, sleep schedule and history of collisions in addition to severity
 - Mostly up to respirologist discretion in non-commercial drivers.
 - “mod-severe OSA not compliant with treatment, who treating physician believes is at increased risk of MVC” should not drive any vehicle (CMA 2023)
 - In commercial drivers with AHI > 20, patients generally need to be “treated adequately” i.e. compliant with therapy
 - Defined as ≥ 4 hrs >70% of nights in past 30 days
- CMA Driver’s Guide 2023 (10th edition) now reflects these changes

Know:



Alert

- If a physician believes that a patient is likely to be at risk while driving because of a symptomatic sleep disorder and the patient refuses investigation by a sleep study or refuses appropriate treatment, that patient should not drive any class of motor vehicle.

Solid Lung Nodules – Fleischner 2017

Guidelines Simplified for IMR

SOLID NODULE	Risk?	<6mm	6-8mm	>8mm
Single Solid Nodule	Low Risk	No routine f/u	CT at 6-12 mos then <i>consider</i> 18-24mos	Consider CT at 3 mos, PET – CT or biopsy
	High Risk	Optional CT at 12 mos	CT at 6-12 mos, AND CT at 18-24 mos.	
Multiple Solid Nodules	Low Risk patient	No routine f/u	CT at 3-6 mos, then <i>consider</i> CT at 18-24 mos*	
	High Risk Patient	Optional CT @ 12mos	CT at 3-6 mos, <i>and</i> CT at 18-24 mos*	

This guideline applies to Adults >35 years and Pertains to incidental nodules, NOT in lung cancer screening population, patients with history of primary cancer, or immunosuppression

*With multiple nodules, use most suspicious nodule to guide interval for follow-up.

High risk = estimated risk of cancer >5%. (use clinical and radiographic risk – older age, heavy smoking, irregular nodule, spiculated, upper lobe location ... there is ambiguity in guideline about this)

Subsolid Lung Nodules – Fleischner 2017

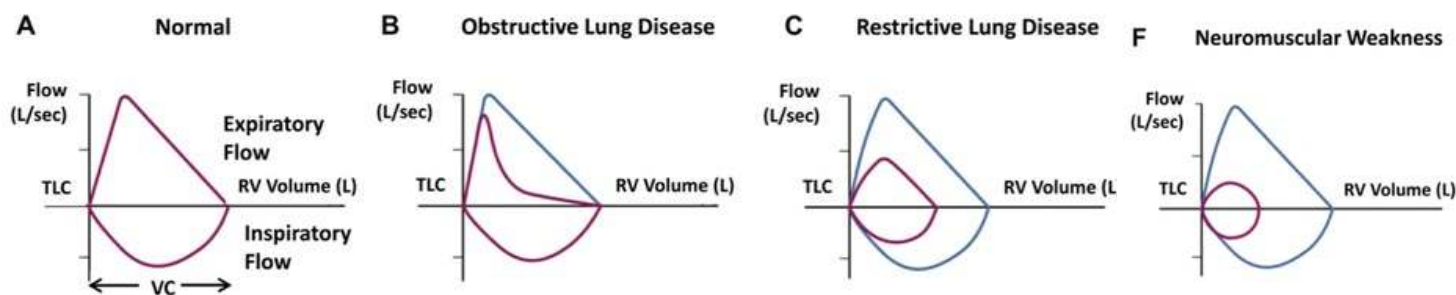
Guidelines Simplified for IMR

SUBSOLID NODULE	Description of nodule	<6mm	>= 6mm	Notes
Single nodule	Ground Glass	No routine f/u	CT at 6-12 mos, then if persistent, every 2y until 5y	Increase in nodule by >=2mm significant
	Partsolid	No routine F/U	CT at 3-6 mos, then if persistent, annual CT to 5 yrs	
Multiple subsolid Nodules	Low risk	CT at 3-6 mos, then if stable consider at 2 and 4 y	CT at 3-6 mos then depends on most suspicious nodule	
	High risk	CT at 3-6 mos then at 2y/4y if high risk		

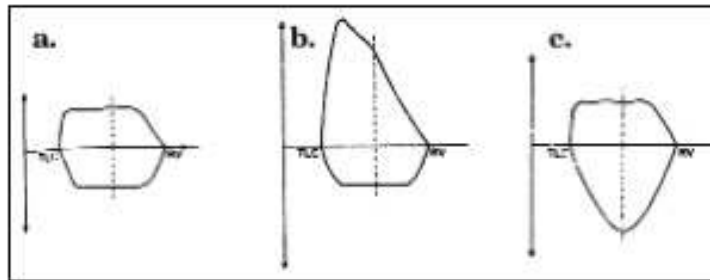
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Spirometry Glossary of Terms

- **FEV1**: Forced Expired Volume in 1 second (in litres)
- **FVC**: Forced Vital Capacity: maximum volume of gas that can be expired when a subject exhales as forcefully and rapidly as possible after maximal inspiration. (in litres).
- **VC**: Vital Capacity (or Slow Vital Capacity): the volume of gas measured from a slow, complete exhalation.
- **Look at patient demographics!**
- **Look at the Curve!** Obstructive vs. restrictive



PFT Approach



- A. Fixed upper-airway obstruction (intrathoracic or extrathoracic)
 - Glottic stenosis (prolonged intubation), subglottic stenosis (GPA, sarcoid, polychondritis)
- B. Variable extrathoracic obstruction
 - Flattening of inspiratory curve
 - Vocal cord dysfunction/paralysis
- C. Variable intrathoracic obstruction
 - Flattening of expiratory curve
 - E.g. tracheomalacia of intrathoracic airway

PFT Approach

Spirometry:

- Is there Obstruction? ($FEV_1/FVC < LLN$)
 - If yes, is it reversible or fixed ($FEV_1 > 12\% + 200cc$)?
 - Is there reduction in FVC suggesting restriction or gas trapping

Volumes:

- Is there restriction? $TLC < LLN$
 - If yes, is it parenchymal (reduced DLCO) or extra parenchymal (normal DLCO)
- Is there gas trapping? $RV/TLC > ULN$
- Is there hyperinflation? $TLC > ULN$

DLCO:

- Is there a diffusion impairment? $DLCO < LLN$ ($DLCO < 75\%$ in some institutions)
 - Isolated decreased DLCO?
 - Increased DLCO?
- Is there concomitant obstructive or restrictive physiology?

PFT Clang Associations and Pitfalls

- Restrictive pattern varying with position
 - Vital Capacity decreases when lying down by >10%
 - Gravity eliminated therefore unmasks diaphragm dysfunction, confirm with MIPs/MEPs
 - May be presented as a post-op scenario (e.g. CABG) or NMD (e.g. ALS)
- Isolated decrease in DLCO?
 - Classically pulmonary hypertension, also early ILD/emphysema, anemia
- Increased DLCO
 - Pulmonary hemorrhage/polycythemia, LV failure, asthma, obesity
- More than one process can occur in the same patient!
 - Mixed obstructive and restrictive process: Obese smoker, bronchiectasis, sarcoidosis, HP
- Ask for full PFTs including lung volumes and DLCO if possible:
 - If “what you see is what you get”, interpret based on the best of your ability:
 - “Mild obstructive lung disease. I would ask for further testing to evaluate for a bronchodilator response and measure lung volumes and DLCO to help with my differential diagnosis.”
- PFTs in OLDER ADULTS / ELDERLY
 - Strong evidence suggest that lung function including FEV1 and FVC ↓ with age. This is why we correct for age in predicted FEV1 and FVCs. Similarly, **FEV1 actually declines faster than FVC so the ratio will also decline with age.**
 - The GOLD cutoff of 0.7 will (intentionally by their reasoning) **overdiagnose COPD** in older patients.
 - ATS suggests using FEV1/FVC Ratio below 5th percentile for adults as obstruction in this setting
 - RV ↑ with age
 - Kyphosis causing decreased height will affect ULN/LLN estimates. Normal range is lower in shorter individuals. Consider using ARM SPAN
- DLCO also declines due to changes in the gas exchange interface.

As a little aside, TLC stays the same throughout life. The drop in FVC is therefore offset by an increase in RV.

PFT Contraindications

- Contraindications to PFTs: (BMJ)
 1. Hemoptysis
 2. Pneumothorax
 3. Unstable cardiovascular status including recent MI
 4. Aneurysms – thoracic, abdominal or cerebral
 5. Recent eye surgery – eg. Cataracts
 6. Recent thoracic or abdominal surgery
 7. Presence of acute illness that may interfere with test performance
- Contraindications to Methacholine Challenge: (ATS)

Absolute:

 1. Severe airflow limitation $FEV_1 < 50\%$ or $< 1L$
 2. Recent MI or Stroke in last 3m
 3. Uncontrolled HTN, SBP $> 200/100$
 4. Known Aortic Aneurysm

Relative:

 1. Moderate airflow limitation $FEV_1 < 60\%$ or $< 1.5L$
 2. Pregnancy or nursing mothers (methacholine is category C)
 3. Use of cholinesterase inhibitor (myasthenia gravis)

PFTs Perioperatively

Patients with lung cancer being considered for surgery should have FEV1 and DLCO measured and **predicted post-op FEV1 and DLCO** should be calculated (ACCP 2013)

- If both **PPO FEV1 and DLCO are >60% predicted** no further testing is needed
 - Low risk for perioperative death and cardiopulmonary complications following resection including pneumonectomy
 - FYI only : predicted postop FEV1 = pre-op FEV1 x (1-y/z)
 - Y is total functional lung segments to be removed (RUL = 3, RML = 2, RLL = 5, LUL = 5, LLL = 4)
 - Z is total number of functional segments
 - If either is <60%, further testing is required (and those are deemed high risk)
 - *Thoracic surgeons do this for a living – GIM would never be involved in determining candidacy for type of lung surgery*
 - Predictors of adverse perioperative pulmonary events:
 1. **Surgical site (aortic> intrathoracic> upper abdominal> abdominal)**
 2. Age, pre-existing lung disease
- Preventing post op complications
- Epidural analgesia decreases atelectasis and pneumonia (Ann Surg 2014)

Diaphragmatic Weakness

- Often idiopathic, often unilateral (and may be entirely asymptomatic)
- Causes include:
 - Trauma or surgery (C3-5 injury during trauma, phrenic nerve injury during cardiac surgery)
 - Mechanical ventilation
 - Myopathy/neuropathy
 - ALS, MG, critical illness myopathy/neuropathy

@IMR we often get asked: **What is the most specific and most sensitive test for diagnosis of diaphragm weakness/paralysis?**

Most SENSITIVE = MIP.

Most SPECIFIC: There isn't really a good **specific** test on PFTs for diaphragmatic weakness. Proceed to ultrasound if you want specificity. If forced to choose we “guess” FVC... because it's associated with poor prognosis in diaphragmatic dysfunction as per the ALS CTS guidelines.



BONUS MCQ 1 – 2026

A 24-year-old woman with depression, asthma, and irritable bowel syndrome with BMI 18 presents to your GIM rapid access clinic with 5-year history of chronic productive cough and wheeze with recurrent episodes of pneumonia treated by multiple walk-in-clinics. CBC shows a normal white count differential. Spirometry was done which showed FEV1/FVC 0.62 FEV1 1.83 L/61% FVC 2.95/74%. She had a CT Chest and it showed a moderate burden of upper lobe predominant bronchiectasis and mucus plugging but is otherwise normal. What is the next best test to establish the most likely diagnosis?

- a) Nasal nitric oxide
- b) A1AT
- c) Sputum AFB
- d) Sweat chloride testing

BONUS MCQ 2 – 2026

A 72-year-old man with a 45-pack-year smoking history presents for follow-up. He has GOLD stage III COPD (FEV₁ 38% predicted) and has had 3 moderate exacerbations in the past year, each requiring oral prednisone and antibiotics. He uses tiotropium daily and a mometasone–formoterol inhaler twice daily. He remains symptomatic with chronic cough and an mMRC dyspnea score of 3. He has no history of asthma, peripheral blood eosinophil count is 30/μL, and his BMI is 22. Prior sputum culture has grown pseudomonas and *M. avium*. Which of the following is the most appropriate next step to reduce his risk of future exacerbations?

- A. Add roflumilast
- B. Add azithromycin
- C. Increase the dose of inhaled corticosteroid
- D. Start dupilumab

BONUS MCQ 3 – 2025

An 82-year-old male with advanced idiopathic pulmonary fibrosis on nintedanib and home oxygen is admitted to hospital worsening dyspnea. He has severe dyspnea at baseline and steady reduction in functional capacity. CT chest is completed in the emergency department showing progression of typical UIP pattern. No evidence of pneumonia, pulmonary embolism, pulmonary edema, or diffuse ground-glass opacities. Patient's oxygen saturation is 88% on 6LPM of supplemental oxygen. Goals of care conversation discussed with patient and they are wanting to focus on quality of life rather than aggressive interventions. What would you recommend next for treatment?

- a) Start oral morphine
- b) Change nintedanib to pirfenidone
- c) Start prednisone
- d) Refer to ICU for consideration of intubation

BONUS MCQ 4 – 2025

A 55-year-old female presents with parotid gland swelling, left facial paralysis, and blurry vision. An ophthalmologist confirms a diagnosis of anterior uveitis and starts her on Prednisone 40mg daily. She reports subjective fevers at home but not currently in your office. CXR show bilateral hilar lymphadenopathy. EBV testing is negative. What would recommend next for investigations?

- a) PFT, ECG, Serum Ca, serum ACE
- b) PFT, ECG, Serum Ca, EBUS
- c) PFT, ECG, Serum Ca, Mediastinoscopy
- d) PFT, ECG, Serum Ca, 1,25 Vit D

BONUS MCQ 5 – 2025

A 27-year-old woman is referred to clinic with progressive exertional dyspnea. She is a never smoker and describes mild dyspnea with occasional pre-syncopal like symptoms. Physical examination remarkable for skin tightening of the fingers and hands along with scattered perioral telangiectasias. The referring physician has attached pulmonary function tests (below). What is the most likely diagnosis?

- a. COPD
- b. Asthma
- c. Shrinking Lung Syndrome
- d. Pulmonary hypertension

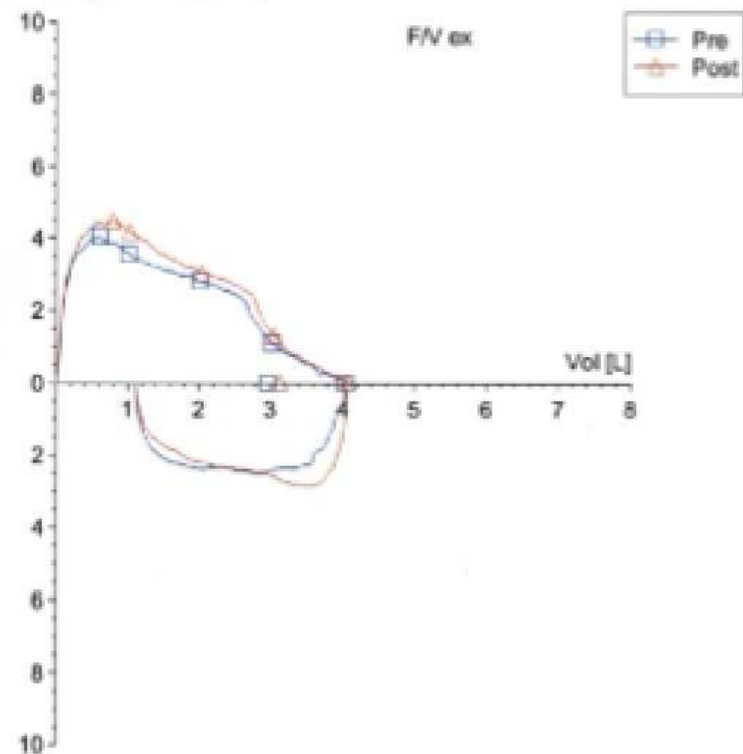
	LLN	ULN	Obs.	%Reference	Post BD
FEV1	2.45	3.87	2.9L	91.3	3.0L
FVC	2.85	4.54	3.3L	89.4	
FEV1/FVC	0.76	0.95	0.88	101.5	
TLC	4.28	6.38	5.2L	98	
RV	0.57	2.01	1.1L	93.8	
DLCO	18.1	28.2	14.3	63	

BONUS MCQ 6 - 2024

A 30y Pregnant woman presents with dyspnea on exertion in pregnancy. She finds she is wheezing when she exercises. Cardiopulmonary exam at rest is entirely normal. Urgent spirometry is obtained and shown. Which of the following is next best step?

- a) Initiate ICS – LABA
- b) Urgent referral to ENT
- c) Initiate Prednisone 1mg/kg
- d) CXR to evaluate for sarcoidosis
- e) c-ANCA

Spirometry Grade (Pre):



BONUS MCQ 7 – 2024

A 30-year-old pregnant woman at 30 weeks gestation presents to the ED with worsening dyspnea and wheezing with new productive sputum. She has had a prior diagnosis of asthma but stopped taking her inhalers during pregnancy. She is febrile but her vitals are otherwise normal, and her ABG shows expected respiratory alkalosis. She is diffusely wheezy on exam. Her chest X-ray suggest an RUL opacity and you suspect a bacterial pneumonia. You decide to hospitalize her for a severe asthma exacerbation. Which of the following would you not recommend at this time?

- a) Restart inhaled budesonide
- b) Schedule q4h and prn short acting bronchodilators
- c) Start oral prednisone 40mg
- d) Start oral Moxifloxacin daily
- e) Consult OB for fetal monitoring

BONUS MCQ 8 2025

A 30-year-old female is referred for a chronic cough. Suspecting asthma, you order spirometry that confirms reversible airflow obstruction. She describes no daytime symptoms but persistent nighttime cough three times a week. She denies activity limitations or recent chest infections. What inhaled therapy would you initiate?

- a) Start standing daily fluticasone-umeclidinium-vilanterol
- b) Start daily ciclesonide and prn salbutamol
- c) Start prn budesonide-formoterol
- d) Start salbutamol prn only

BONUS MCQ 9 2025

In your clinic, you see a 72-year-old male three weeks after a recent hospital admission for an infectious exacerbation of COPD. He is currently on tiotropium bromide monotherapy daily and scores 12 on his CAT. He was discharged home on room air with an ABG showing a pCO₂ of 50 and pO₂ of 62. Which of the following intervention have a mortality benefit?

- a) Initiate home oxygen
- b) Referral for initiation of home NIV
- c) Change inhaler therapy to daily tiotropium-olodaterol
- d) Change inhaler therapy to daily fluticasone-umeclidinium-vilanterol

BONUS MCQ 10 (2025)

A 53-year-old male is admitted to the medical ward with a 5-day history of productive cough, chest pain, and exertional dyspnea. Point-of-care ultrasound demonstrates large right-sided pleural effusion. You perform a diagnostic thoracentesis. For which of the following fluid compositions would you **not** recommend insertion of a chest drain?

- a) pH 7.15, LDH 300, culture and gram stain no growth/no organisms seen.
- b) pH 7.25, LDH 1100, culture and gram stain no growth/no organisms seen.
- c) pH not available, glucose 3.5, LDH 800 culture and gram stain no growth/no organisms seen
- d) pH 7.30, LDH 300, gram positive cocci in chains on gram stain

BONUS MCQ 11 (2025)

A 21-year-old male presents to the ED given abnormal imaging results. As part of required TB screening at work, he had a CXR demonstrating a 3.5cm right apical pneumothorax. He is asymptomatic and is a non-smoker. His vitals are normal, and your exam is unremarkable except for decreased air entry in right apex. You would now recommend:

- a) Insertion of a right small bore chest tube and admit to hospital
- b) Insertion of ambulatory pneumothorax device and discharge with close follow-up
- c) Needle aspiration and then discharge with close follow-up
- d) Discharge home with close follow up

BONUS MCQ Answers

- 1) D – upper lobe bronchiectasis with low BMI raises concern for cystic fibrosis. There are cases which are missed until diagnosed in adulthood. PCD presents with symptoms since early childhood and organ laterality defects like situs inversus which would be seen on CT. A1AT would present with concurrent emphysema. TB/NTM is upper lobe predominant but would also usually have other parenchymal changes (nodules, cavities, etc).
- 2) A – roflumilast to reduce exacerbations in chronic bronchitis phenotype. Chronic azithromycin is avoided in patients with NTM in case the NTM will be considered for treatment in the future. The patient's peripheral eosinophils are not high enough to meet criteria for consideration of dupilumab.
- 3) A – Progressive IPF with functional decline and limiting dyspnea. Palliative management of dyspnea with symptom benefit from low dose oral opioids. No role for prednisone (harm on chronic basis – PANTHER – and limited data for acute, patient doesn't have a IPF flare).
- 4) D– This is Heerfordt syndrome / sarcoid - Do not get an ACE level. It is not helpful with the diagnosis. Some sarcoid experts may break this rule but not useful to a generalist. Heerfordt syndrome does not require a tissue diagnosis. Avoid unnecessary interventions.
- 5) D – isolated reduction in DLCO ddx includes pulmonary HTN, early ILD, or anemia. Patient likely has limited scleroderma which is highly associated with pulmonary HTN.
- 6) B – Send to ENT! Real case from Dr McKenna office – pt sent directly to my ENT husband same day who confirmed subglottic stenosis (subsequent workup incl C ANCA, IgG4 etc all negative = idiopathic).
- 7) D - Steroids are recommended to be used in asthma exacerbations during pregnancy as an untreated asthma exacerbation in pregnancy poses significant risks to mother and baby. quinolones are recommended against in pregnancy especially when other safe alternatives apply here - In this case, likely choose a beta lactam for her CAP.

BONUS MCQ Answers

- 8) B - Follow the CTS guidelines here. Frequent nighttime symptoms (>1/week) indicate poorly controlled asthma. Start with daily ICS. GINA and some experts would suggest c is an acceptable alternative.
- 9) D - Based on analysis of IMPACT and ETHOS. Recent severe exacerbation with CAT >10. Meets indication for **triple therapy with mortality benefit**. PCO₂>52 is required for home NIV, pO₂ too high. No established mortality benefit with (c)
(d) would also be next best step for management based on prompt. Of note PULMONARY REHAB would also have a mortality benefit if started within the next week (<4 weeks post AECOPD discharge)
- 10) C - **pH not available, glucose 3.5, LDH 800 culture and gram stain no growth/no organisms seen**
- 11) D - No intervention required for asymptomatic pneumothorax regardless of size. Do not discharge home without appropriate follow-up! May consider short period of observation in ED.

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